20

The First Law of Thermodynamics

CHAPTER OUTLINE

20.1 Heat and Internal Energy

20.2 Specific Heat and Calorimetry

20.3 Latent Heat

20.4 Work and Heat in Thermodynamic Processes

20.5 The First Law of Thermodynamics

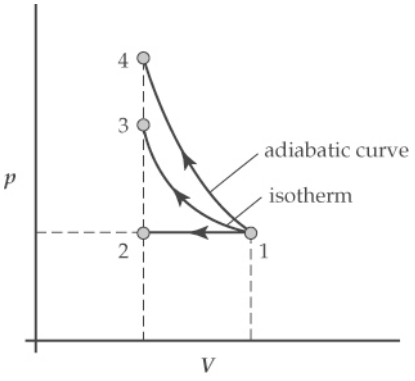
20.6 Some Applications of the First Law of Thermodynamics

20.7 Energy Transfer Mechanisms in Thermal Processes

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ20.1 Answer (b). The work done on a gas equals the area under the process curve in a PV diagram. In an isobaric process, the pressure is constant, so $P_f = P_i$ and the work done is the area under curve 1–2 in ANS. FIG. OQ20.1. For an isothermal process, the ideal gas law gives $P_f V_f = P_i V_i$, so $P_f = (V_i/V_f)\, P_i = 2P_i$ and the work done is the area under curve 1–3 in ANS. FIG. OQ20.1. For an adiabatic process, $P_f V_f^{\gamma} = P_i V_i^{\gamma} =$ constant (see Ch. 21), so $P_f = (V_i/V_f)^{\gamma} P_i$ and $P_f = 2^{\gamma} P_i > 2P_i$ since $\gamma > 1$ for all ideal gases. The work done in an adiabatic process is the area under curve 1–4, which exceeds that done in either of the other processes.



ANS. FIG. OQ20.1

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OQ20.2 Answer (d). The high specific heat will keep the end in the fire from warming up very fast. The low conductivity will make the handle end warm up only very slowly.

OQ20.3 Answer (a). Do a few trials with water at different original temperatures and choose the one where room temperature is halfway between the original and the final temperature of the water. Then you can reasonably assume that the contents of the calorimeter gained and lost equal quantities of heat to the surroundings, for net transfer zero. James Joule did it like this in his basement in London.

OQ20.4 Answer (c). Since less energy was required to produce a 5°C rise in the temperature of the ice than was required to produce a 5°C rise in temperature of an equal mass of water, we conclude that the specific heat of ice $\left[c = Q / m(\Delta T)\right]$ is less than that of water.

OQ20.5 Answer (e). The required energy input is

$$Q = mc(\Delta T) = (5.00\ \text{kg})(128\ \text{J/kg}\cdot\text{°C})(327\text{°C} - 20.0\text{°C})$$
$$= 1.96 \times 10^5\ \text{J}$$

OQ20.6 Answer (c). With a specific heat half as large, the ΔT is twice as great in the ethyl alcohol.

OQ20.7 Answer (d). From the relation $Q = mc\Delta T$, the change in temperature of a substance depends on the quantity of energy Q added to that substance, and its specific heat and mass: $\Delta T = Q/mc$. The masses of the substances are not given.

OQ20.8 Rankings (e) > (a) = (b) = (c) > (d). We think of the product $mc\Delta T$ in each case, with c = 1 for water and about 0.5 for beryllium: (a) $1 \cdot 1 \cdot 6 = 6$, (b) $2 \cdot 1 \cdot 3 = 6$, (c) $2 \cdot 1 \cdot 3 = 6$, (d) $2(0.5)3 = 3$, (e) > 6 because a large quantity of energy input is required to melt the ice.

OQ20.9 (i) Answer (d). (ii) Answer (d). Internal energy and temperature both increase by minuscule amounts due to the work input.

OQ20.10 Answer (b). The total change in internal energy is zero.

$$Q_{\text{Cu}} + Q_{\text{water}} + Q_{\text{Al}} = 0$$
$$(100\ \text{g})\left(0.092\frac{\text{cal}}{\text{g °C}}\right)(T_f - 95.0\text{°C})$$
$$+(200\ \text{g})\left(1.00\frac{\text{cal}}{\text{g °C}}\right)(T_f - 15.0\text{°C})$$
$$+(280\ \text{g})\left(0.215\frac{\text{cal}}{\text{g °C}}\right)(T_f - 15.0\text{°C}) = 0$$

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$$9.20T_f - 874°\text{C} + 200T_f - 3\ 000°\text{C} + 60.2T_f - 903°\text{C} = 0$$
$$269.4T_f = 4\ 777°\text{C}$$
$$T_f = 17.7°\text{C}$$

OQ20.11 Answer (e). Twice the radius means four times the surface area. Twice the absolute temperature makes T^4 sixteen times larger in Stefan's law. The total effect is $4 \times 16 = 64$.

OQ20.12 Answer (d). During isothermal compression, the temperature remains unchanged. The internal energy of an ideal gas is proportional to its absolute temperature. As the gas is compressed, positive work is done on the gas but also energy is transferred from the gas by heat because the total change in internal energy is zero.

OQ20.13 Answer (c) only. By definition, in an adiabatic process, no energy is transferred to or from the gas by heat. In an expansion process, the gas does work on the environment. Since there is no energy input by heat, the first law of thermodynamics says that the internal energy of the ideal gas must decrease, meaning the temperature will decrease. Also, in an adiabatic process, PV^γ = constant, meaning that the pressure must decrease as the volume increases.

OQ20.14 Answer (b) only. In an isobaric process on an ideal gas, pressure is constant while the gas either expands or is compressed. Since the volume of the gas is changing, work is done either on or by the gas. Also, from the ideal gas law with pressure constant, $P\Delta V = nR\Delta T$; thus, the gas must undergo a change in temperature having the same sign as the change in volume. If $\Delta V > 0$, then both ΔT and the change in the internal energy of the gas are positive ($\Delta U > 0$). However, when $\Delta V > 0$, the work done *on the gas* is negative ($\Delta W < 0$), and the first law of thermodynamics says that there must be a positive transfer of energy by heat to the gas ($Q = \Delta U - W > 0$). When $\Delta V < 0$, a similar argument shows that $\Delta U < 0$, $W > 0$, and $Q = \Delta U - W < 0$. Thus, all of the other listed choices are false statements.

OQ20.15 Answer (d). The temperature of the ice must be raised to the melting point, $\Delta T = +20.0°\text{C}$, before it will start to melt. The total energy input required to melt the 1.00 kg of ice is

$$Q = mc_{\text{ice}}(\Delta T) + mL_f = (1.00\ \text{kg})\left[(2\ 090\ \text{J/kg}\cdot°\text{C})(20.0°\text{C}) + 3.33\times10^5\ \text{J/kg}\right] = 3.75\times10^5\ \text{J}$$

The time the heating element will need to supply this quantity of energy is

$$\Delta t = \frac{Q}{P} = \frac{3.75\times10^5\ \text{J}}{1.00\times10^3\ \text{J/s}} = (375\ \text{s})\left(\frac{1\ \text{min}}{60\ \text{s}}\right) = 6.25\ \text{min}$$

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ANSWERS TO CONCEPTUAL QUESTIONS

CQ20.1 Rubbing a surface results in friction converting kinetic energy to thermal energy. Metal, being a good thermal conductor, allows energy to transfer swiftly out of the rubbed area to the surrounding areas, resulting in a swift fall in temperature. Wood, being a poor conductor, permits a slower rate of transfer, so the temperature of the rubbed area does not fall as swiftly.

CQ20.2 Keep them dry. The air pockets in the pad conduct energy by heat, but only slowly. Wet pads would absorb some energy in warming up themselves, but the pot would still be hot and the water would quickly conduct a lot of energy right into you.

CQ20.3 Heat is a method of transferring energy, not energy contained in an object. Further, a low-temperature object with large mass, or an object made of a material with high specific heat, can contain more internal energy than a higher-temperature object.

CQ20.4 There are three properties to consider here: thermal conductivity, specific heat, and mass. With dry aluminum, the thermal conductivity of aluminum is much greater than that of (dry) skin. This means that the internal energy in the aluminum can more readily be transferred to the atmosphere than to your fingers. In essence, your skin acts as a thermal insulator. If the aluminum is wet, it can wet the outer layer of your skin to make it into a good thermal conductor; then more energy from the aluminum can transfer to you. Further, the water itself, with additional mass and with a relatively large specific heat compared to aluminum, can be a significant source of extra energy to burn you. In practical terms, when you let go of a hot, dry piece of aluminum foil, the energy transfer by heat immediately ends. When you let go of a hot *and* wet piece of aluminum foil, the hot water sticks to your skin, continuing the heat transfer, and resulting in more energy transfer by heat to you!

CQ20.5 If the system is isolated, no energy enters or leaves the system by heat, work, or other transfer processes. Within the system energy can change from one form to another, but since energy is conserved these transformations cannot affect the total amount of energy. The total energy is constant.

CQ20.6 (a) Warm a pot of coffee on a hot stove.

(b) Place an ice cube at 0°C in warm water—the ice will absorb energy while melting, but not increase in temperature.

(c) Let a high-pressure gas at room temperature slowly expand by pushing on a piston. Energy comes out of the gas by work in a

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constant-temperature expansion as the same quantity of energy flows by heat in from the surroundings.

(d) Warm your hands by rubbing them together. Heat your tepid coffee in a microwave oven. Energy input by work, by electromagnetic radiation, or by other means, can all alike produce a temperature increase.

(e) Davy's experiment is an example of this process.

CQ20.7 (a) Yes, wrap the blanket around the ice chest. The environment is warmer than the ice, so the blanket prevents energy transfer by heat from the environment to the ice.

(b) Explain to your little sister that her body is warmer than the environment and requires energy transfer by heat into the air to remain at a fixed temperature. The blanket will prevent this conduction and cause her to feel warmer, not cool like the ice.

CQ20.8 Ice is a poor thermal conductor, and it has a high specific heat. The idea behind wetting fruit is that a coating of ice prevents the fruit from cooling below the freezing temperature even as the air outside is colder, and also to protect plants from frost. When frost melts it takes its heat from the fruit, and kills it. When ice melts it takes heat from the air, so it acts as insulation for the fruit.

CQ20.9 The person should add the cream immediately when the coffee is poured. Then the smaller temperature difference between coffee and environment will reduce the rate of energy transfer out of the cup during the several minutes.

CQ20.10 The sunlight hitting the peaks warms the air immediately around them. This air, which is slightly warmer and less dense than the surrounding air, rises, as it is buoyed up by cooler air from the valley below. The air from the valley flows up toward the sunny peaks, creating the morning breeze.

CQ20.11 Because water has a high specific heat, it can absorb or lose quite a bit of energy and not experience much change in temperature. The water would act as a means of preventing the temperature in the cellar from varying much so that stored goods would neither freeze nor become too warm.

CQ20.12 The steam locomotive engine is one perfect example of turning internal energy into mechanical energy. Liquid water is heated past the point of vaporization. Through a controlled mechanical process, the expanding water vapor is allowed to push a piston. The translational kinetic energy of the piston is usually turned into rotational kinetic energy of the drive wheel.

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SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 20.1 Heat and Internal Energy

P20.1 (a) The energy equivalent of 540 Calories is found from

$$Q = 540\ \cancel{\text{Cal}}\left(\frac{10^3\ \cancel{\text{cal}}}{1\ \cancel{\text{Cal}}}\right)\left(\frac{4.186\ \text{J}}{1\ \cancel{\text{cal}}}\right) = \boxed{2.26\times10^6\ \text{J}}$$

(b) The work done lifting her weight mg up one stair of height h is $W_1 = mgh$. Thus, the total work done in climbing N stairs is $W = Nmgh$, and we have $Q = Nmgh$, or

$$N = \frac{Q}{mgh} = \frac{2.26\times10^6\ \text{J}}{(55.0\ \text{kg})(9.80\ \text{m/s}^2)(0.150\ \text{m})} \simeq \boxed{2.80\times10^4\ \text{stairs}}$$

(c) If only 25% of the energy from the donut goes into mechanical energy, we have

$$N = \frac{0.25Q}{mgh} = 0.25\left(\frac{Q}{mgh}\right) = 0.25(2.80\times10^4\ \text{stairs})$$

$$= \boxed{6.99\times10^3\ \text{stairs}}$$

Section 17.2 Specific Heat and Calorimetry

P20.2 The container is thermally insulated, so no energy is transferred by heat:

$$Q = 0$$

and $$\Delta E_{\text{int}} = Q + W_{\text{input}} = 0 + W_{\text{input}} = 2mgh$$

The work on the falling weights is equal to the work done on the water in the container by the rotating blades. This work results in an increase in internal energy of the water:

$$2mgh = \Delta E_{\text{int}} = m_{\text{water}}c\Delta T$$

$$\Delta T = \frac{2mgh}{m_{\text{water}}c} = \frac{2(1.50\ \text{kg})(9.80\ \text{m/s}^2)(3.00\ \text{m})}{(0.200\ \text{kg})(4\,186\ \text{J/kg}\cdot\text{°C})} = \frac{88.2\ \text{J}}{837\ \text{J/°C}}$$

$$= \boxed{0.105\text{°C}}$$

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P20.3 The system is thermally isolated, so

$$Q_{\text{water}} + Q_{\text{Al}} + Q_{\text{Cu}} = 0$$

$$(0.250\ \text{kg})\left(4\,186\frac{\text{J}}{\text{kg}\cdot^\circ\text{C}}\right)(T_f - 20.0^\circ\text{C}) + (0.400\ \text{kg})\left(900\frac{\text{J}}{\text{kg}\cdot^\circ\text{C}}\right)(T_f - 26.0^\circ\text{C}) + (0.100\ \text{kg})\left(387\frac{\text{J}}{\text{kg}\cdot^\circ\text{C}}\right)(T_f - 100^\circ\text{C}) = 0$$

$$1\,046.5T_f - 20\,930^\circ\text{C} + 360T_f - 9\,360^\circ\text{C} + 38.7T_f - 3\,870^\circ\text{C} = 0$$

$$1\,445.2T_f = 34\,160^\circ\text{C}$$

$$T_f = \boxed{23.6^\circ\text{C}}$$

P20.4 As mass m of water drops from top to bottom of the falls, the gravitational potential energy given up (and hence, the kinetic energy gained) is $Q = mgh$. If all of this goes into raising the temperature, $Q = mc\Delta T$, and the rise in temperature will be

$$\Delta T = \frac{Q}{mc_{\text{water}}} = \frac{\cancel{m}gh}{\cancel{m}c_{\text{water}}} = \frac{(9.80\ \text{m/s}^2)(807\ \text{m})}{4\,186\ \text{J/kg}\cdot^\circ\text{C}} = 1.89^\circ\text{C}$$

and the final temperature is

$$T_f = T_i + \Delta T = 15.0^\circ\text{C} + 1.89^\circ\text{C} = \boxed{16.9^\circ\text{C}}$$

P20.5 When thermal equilibrium is reached, the water and aluminum will have a common temperature of $T_f = 65.0^\circ\text{C}$. Assuming that the water-aluminum system is thermally isolated from the environment, $Q_{\text{cold}} = -Q_{\text{hot}}$:

$$m_w c_w (T_f - T_{i,w}) = -m_{\text{Al}} c_{\text{Al}} (T_f - T_{i,\text{Al}})$$

or

$$m_w = \frac{-m_{\text{Al}} c_{\text{Al}} (T_f - T_{i,\text{Al}})}{c_w (T_f - T_{i,w})} = \frac{-(1.85\ \text{kg})(900\ \text{J/kg}\cdot^\circ\text{C})(65.0^\circ\text{C} - 150^\circ\text{C})}{(4\,186\ \text{J/kg}\cdot^\circ\text{C})(65.0^\circ\text{C} - 25.0^\circ\text{C})} = \boxed{0.845\ \text{kg}}$$

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P20.6 We find its specific heat from the definition, which is contained in the equation $Q = mc_{\text{silver}}\Delta T$ for energy input by heat to produce a temperature change. Solving, we have

$$c_{\text{silver}} = \frac{Q}{m\Delta T}$$

$$c_{\text{silver}} = \frac{1.23\times10^{3}\ \text{J}}{(0.525\ \text{kg})(10.0°\text{C})} = \boxed{234\ \text{J/kg}\cdot°\text{C}}$$

P20.7 We imagine the stone energy reservoir has a large area in contact with air and is always at nearly the same temperature as the air. Its overnight loss of energy is described by

$$P = \frac{Q}{\Delta t} = \frac{mc\Delta T}{\Delta t}$$

$$m = \frac{P\Delta t}{c\Delta T} = \frac{(-6\,000\ \text{J/s})(14\ \text{h})(3\,600\ \text{s/h})}{(850\ \text{J/kg}\cdot°\text{C})(18.0°\text{C} - 38.0°\text{C})}$$

$$= \frac{3.02\times10^{8}\ \text{J}}{(850\ \text{J/kg}\cdot°\text{C})(20.0°\text{C})} = \boxed{1.78\times10^{4}\ \text{kg}}$$

***P20.8** From $Q = mc\Delta T$ we find

$$\Delta T = \frac{Q}{mc} = \frac{1\,200\ \text{J}}{(0.050\,0\ \text{kg})(387\ \text{J/kg}\cdot°\text{C})} = 62.0°\text{C}$$

Thus, the final temperature is $25.0°\text{C} + 62.0°\text{C} = \boxed{87.0°\text{C}}$.

P20.9 Let us find the energy transferred in one minute:

$$Q = \left[m_{\text{cup}}c_{\text{cup}} + m_{\text{water}}c_{\text{water}}\right]\Delta T$$

$$Q = \left[(0.200\ \text{kg})(900\ \text{J/kg}\cdot°\text{C}) + (0.800\ \text{kg})(4\,186\ \text{J/kg}\cdot°\text{C})\right]$$
$$\times(-1.50°\text{C}) = -5\,290\ \text{J}$$

If this much energy is removed from the system each minute, the rate of removal is

$$P = \frac{|Q|}{\Delta t} = \frac{5\,290\ \text{J}}{60.0\ \text{s}} = 88.2\ \text{J/s} = \boxed{88.2\ \text{W}}$$

P20.10 We use $Q_{\text{cold}} = -Q_{\text{hot}}$ to find the equilibrium temperature:

$$m_{\text{Al}}c_{\text{Al}}(T_f - T_c) + m_c c_w(T_f - T_c) = -m_h c_w(T_f - T_h)$$

$$(m_{\text{Al}}c_{\text{Al}} + m_c c_w)T_f - (m_{\text{Al}}c_{\text{Al}} + m_c c_w)T_c = -m_h c_w T_f + m_h c_w T_h$$

$$(m_{\text{Al}}c_{\text{Al}} + m_c c_w + m_h c_w)T_f = (m_{\text{Al}}c_{\text{Al}} + m_c c_w)T_c + m_h c_w T_h$$

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solving for the final temperature gives

$$T_f = \boxed{\frac{(m_{\text{Al}}c_{\text{Al}} + m_c c_w)T_c + m_h c_w T_h}{m_{\text{Al}}c_{\text{Al}} + m_c c_w + m_h c_w}}$$

P20.11 We assume that the water-horseshoe system is thermally isolated (insulated) from the environment for the short time required for the horseshoe to cool off and the water to warm up. Then the total energy input from the surroundings is zero, as expressed by $Q_{\text{Fe}} + Q_{\text{water}} = 0$:

$$(mc\Delta T)_{\text{Fe}} + (mc\Delta T)_{\text{water}} = 0$$

$$m_{\text{Fe}}c_{\text{Fe}}(T - 600°\text{C}) + m_w c_w (T - 25.0°\text{C}) = 0$$

Note that the first term in this equation is a negative number of joules, representing energy lost by the originally hot subsystem, and the second term is a positive number with the same absolute value, representing energy gained by heat by the cold stuff. Solving for the final temperature gives

$$T = \frac{m_w c_w (25.0^\circ\text{C}) + m_{\text{Fe}}c_{\text{Fe}}(600^\circ\text{C})}{m_{\text{Fe}}c_{\text{Fe}} + m_w c_w}$$

Substituting $c_w = 4\,186\ \text{J/kg} \cdot °\text{C}$ and $c_{\text{Fe}} = 448\ \text{J/kg} \cdot °\text{C}$ and suppressing units, we obtain

$$T = \frac{(20.0)(4\,186)(25.0°\text{C}) + (1.50)(448)(600°\text{C})}{(1.50)(448) + (20.0\ \text{kg})(4\,186)}$$

$$= \boxed{29.6°\text{C}}$$

P20.12 (a) The work that the bit does in deforming the block, breaking chips off, and giving them kinetic energy is not a final destination for energy. All of this work turns entirely into internal energy as soon as the chips stop their macroscopic motion. The amount of energy input to the steel is the work done by the bit:

$$W = \vec{\mathbf{F}} \cdot \Delta\vec{\mathbf{r}} = (3.20\ \text{N})(40.0\ \text{m/s})(15.0\ \text{s})\cos 0.00° = 1\,920\ \text{J}$$

To evaluate the temperature change produced by this energy we imagine injecting the same quantity of energy as heat from a stove. The bit, chips, and block all undergo the same temperature change. Any difference in temperature between one bit of steel and another would erase itself by causing an energy transfer by heat from the temporarily hotter to the colder region.

$$Q = mc\Delta T$$

$$\Delta T = \frac{Q}{mc} = \frac{1\,920\ \text{J}}{(0.267\ \text{kg})(448\ \text{J/kg} \cdot °\text{C})} = \boxed{16.1°\text{C}}$$

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(b) See part (a). The same amount of work is done. $\boxed{16.1°\text{C}}$

(c) It makes no difference whether the drill bit is dull or sharp, or how far into the block it cuts. The answers to (a) and (b) are the same because all of the work done by the bit on the block constitutes energy being transferred into the internal energy of the steel.

P20.13 (a) To find the specific heat of the unknown sample, we start with $Q_{\text{cold}} = -Q_{\text{hot}}$ and substitute:

$$(m_w c_w + m_c c_c)(T_f - T_c) = -m_{\text{Cu}} c_{\text{Cu}} (T_f - T_{\text{Cu}}) - m_{\text{unk}} c_{\text{unk}} (T_f - T_{\text{unk}})$$

where w is for water, c the calorimeter, Cu the copper sample, and "unk" the unknown.

$$\left[(0.250\ \text{kg})(4\,186\ \text{J/kg}\cdot°\text{C}) + (0.100\ \text{kg})(900\ \text{J/kg}\cdot°\text{C})\right](20.0°\text{C} - 10.0°\text{C})$$
$$= -(0.050\ 0\ \text{kg})(387\ \text{J/kg}\cdot°\text{C})(20.0 - 80.0)°\text{C} - (0.070\ 0\ \text{kg})c_{\text{unk}}(20.0°\text{C} - 100°\text{C})$$
$$1.020\ 4 \times 10^4\ \text{J} = (5.60\ \text{kg}\cdot°\text{C})c_{\text{unk}}$$
$$c_{\text{unk}} = \boxed{1.82 \times 10^3\ \text{J/kg}\cdot°\text{C}}$$

(b) We cannot make definite identification. It might be beryllium.

(c) The material might be an unknown alloy or a material not listed in the table.

P20.14 (a) Expressing the percentage change as $f = 0.60$, we have

$$(f)(mgh) = mc\Delta T \quad \rightarrow \quad \Delta T = \frac{fgh}{c}$$
$$\Delta T = \frac{(0.600)(9.80\ \text{m/s}^2)(50.0\ \text{m})}{387\ \text{J/kg}\cdot°\text{C}} = 0.760°\text{C} = T - 25.0°\text{C}$$

which gives $\boxed{T = 25.8°\text{C}}$

(b) As shown above, the symbolic result from part (a) shows no dependence on mass. Both the change in gravitational potential energy and the change in internal energy of the system depend on the mass, so the mass cancels.

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P20.15 (a) The gas comes to an equilibrium temperature according to

$$(mc\Delta T)_{\text{cold}} = -(mc\Delta T)_{\text{hot}}$$
$$n_1 Mc\left(T_f - 300\text{ K}\right) + n_2 Mc\left(T_f - 450\text{ K}\right) = 0$$

The molar mass M and specific heat divide out, and we can express n in terms of P, V, and T, using $PV = nRT$:

$$\frac{P_1V_1}{T_1}\cancel{Mc}\left(T_f - T_1\right) + \frac{P_2V_2}{T_2}\cancel{Mc}\left(T_f - T_2\right) = 0$$
$$\frac{P_1V_1}{T_1}T_f - \frac{P_1V_1}{T_1}T_1 + \frac{P_2V_2}{T_2}T_f - \frac{P_2V_2}{T_2}T_2 = 0$$
$$T_f\left(\frac{P_1V_1}{T_1} + \frac{P_2V_2}{T_2}\right) = P_1V_1 + P_2V_2$$

$$T_f = \frac{P_1V_1 + P_2V_2}{\left(\frac{P_1V_1}{T_1} + \frac{P_2V_2}{T_2}\right)} = \frac{(1.75\text{ atm})(16.8\text{ L}) + (2.25\text{ atm})(22.4\text{ L})}{\left(\frac{(1.75\text{ atm})(16.8\text{ L})}{300\text{ K}} + \frac{(2.25\text{ atm})(22.4\text{ L})}{450\text{ K}}\right)}$$
$$= \boxed{380\text{ K}}$$

(b) The pressure of the whole sample in its final state is

$$P_f = (n_1 + n_2)\frac{R}{V_f}T_f = \left(\frac{P_1V_1}{RT_1} + \frac{P_2V_2}{RT_2}\right)\left(\frac{R}{V_1 + V_2}\right)\frac{P_1V_1 + P_2V_2}{\left(\frac{P_1V_1}{T_1} + \frac{P_2V_2}{T_2}\right)}$$

$$P_f = \left(\frac{P_1V_1 + P_2V_2}{V_1 + V_2}\right) = \left(\frac{(1.75\text{ atm})(16.8\text{ L}) + (2.25\text{ atm})(22.4\text{ L})}{16.8\text{ L} + 22.4\text{ L}}\right)$$
$$= \boxed{2.04\text{ atm}}$$

Section 20.3 Latent Heat

***P20.16** To find the amount of steam to be condensed, we begin with

$$Q_{\text{cold}} = -Q_{\text{hot}}$$

With the steam at 100°C, this becomes

$$(m_w c_w + m_c c_c)\left(T_f - T_i\right) = -m_s\left[-L_v + c_w\left(T_f - 100\right)\right]$$

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Substituting numerical values,

$$[(0.250\text{ kg})(4\,186\text{ J/kg}\cdot°\text{C})+(0.050\,0\text{ kg})(387\text{ J/kg}\cdot°\text{C})](50.0°\text{C}-20.0°\text{C})$$
$$=-m_s\left[-2.26\times10^6\text{ J/kg}+(4\,186\text{ J/kg}\cdot°\text{C})(50.0°\text{C}-100°\text{C})\right]$$

Solving for the mass of steam gives

$$m_s=\frac{3.20\times10^4\text{ J}}{2.47\times10^6\text{ J/kg}}=0.012\,9\text{ kg}=\boxed{12.9\text{ g steam}}$$

***20.17** We assume that all work done against friction is used to melt the snow. Equation 8.2 for conservation of energy then gives

$$W_{\text{skier}}=Q_{\text{snow}}$$

or $$f\cdot d=m_{\text{snow}}L_f$$

where $$f=\mu_k n=\mu_k\left(m_{\text{skier}}g\right)$$

Substituting and solving for the distance gives

$$d=\frac{m_{\text{snow}}L_f}{\mu_k\left(m_{\text{skier}}g\right)}=\frac{(1.00\text{ kg})(3.33\times10^5\text{ J/kg})}{0.200(75.0\text{ kg})(9.80\text{ m/s}^2)}$$
$$=2.27\times10^3\text{ m}=\boxed{2.27\text{ km}}$$

P20.18 The energy input needed is the sum of the following terms:

$$Q_{\text{needed}}=(\text{energy to reach melting point})+(\text{energy to melt})+(\text{energy to reach boiling point})+(\text{energy to vaporize})+(\text{energy to reach }110°\text{C})$$

Thus, we have

$$Q_{\text{needed}}=(0.040\,0\text{ kg})\left[(2\,090\text{ J/kg}\cdot°\text{C})(10.0°\text{C})+(3.33\times10^5\text{ J/kg})+(4\,186\text{ J/kg}\cdot°\text{C})(100°\text{C})+(2.26\times10^6\text{ J/kg})+(2\,010\text{ J/kg}\cdot°\text{C})(10.0°\text{C})\right]$$

$$Q_{\text{needed}}=\boxed{1.22\times10^5\text{ J}}$$

P20.19 Remember that energy must be supplied to melt the ice before its temperature will begin to rise. Then, assuming a thermally isolated system, $Q_{\text{cold}}=-Q_{\text{hot}}$, or

$$m_{\text{ice}}L_f+m_{\text{ice}}c_{\text{water}}\left(T_f-0°\text{C}\right)=-m_w c_{\text{water}}\left(T_f-25°\text{C}\right)$$

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and

$$T_f = \frac{m_w c_{\text{water}}(25°\text{C}) - m_{\text{ice}} L_f}{(m_{\text{ice}} + m_w) c_{\text{water}}}$$

$$= \frac{(825\ \cancel{g})(4\,186\ \text{J/kg}\cdot°\text{C})(25°\text{C}) - (75\ \cancel{g})(3.33\times10^5\ \text{J/kg})}{(75\ \cancel{g} + 825\ \cancel{g})(4\,186\ \text{J/kg}\cdot°\text{C})}$$

yielding $\boxed{T_f = 16.3°\text{C}}$

P20.20 The bullet will not melt all the ice, so its final temperature is 0°C.

Then, conservation of energy gives

$$\left(\frac{1}{2}mv^2 + mc|\Delta T|\right)_{\text{bullet}} = m_w L_f$$

where m_w is the mass of melted ice. Solving for m_w gives,

$$m_w = \left(\frac{3.00\times10^{-3}\ \text{kg}}{3.33\times10^5\ \text{J/kg}}\right) \times\left[(0.500)(240\ \text{m/s})^2 + (128\ \text{J/kg}\cdot°\text{C})(30.0°\text{C})\right]$$

$$m_w = \frac{86.4\ \text{J} + 11.5\ \text{J}}{333\,000\ \text{J/kg}} = \boxed{0.294\ \text{g}}$$

P20.21 (a) With 10.0 g of steam added to 50.0 g of ice, we first compute the energy required to melt all the ice:

$$Q_1 = (\text{energy to melt all the ice})$$

$$= (50.0\times10^{-3}\ \text{kg})(3.33\times10^5\ \text{J/kg}) = 1.67\times10^4\ \text{J}$$

Also, the energy required to raise the temperature of the melted ice to 100°C is

$$Q_2 = (\text{energy to raise temp of ice to } 100°\text{C})$$

$$= (50.0\times10^{-3}\ \text{kg})(4\,186\ \text{J/kg}\cdot°\text{C})(100°\text{C}) = 2.09\times10^4\ \text{J}$$

Thus, the total energy to melt all of the ice and raise its temperature to 100°C is

$$Q_1 + Q_2 = 1.67\times10^4\ \text{J} + 2.09\times10^4\ \text{J} = 3.76\times10^4\ \text{J}$$

The energy available from the condensation of 10.0 g of steam is

$$Q_3 = (\text{energy available as steam condenses})$$

$$= (10.0\times10^{-3}\ \text{kg})(2.26\times10^6\ \text{J/kg}) = 2.26\times10^4\ \text{J}$$

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Thus, we see that $Q_3 > Q_1$, but $Q_3 < Q_1 + Q_2$, which means that all of the ice will melt, $\boxed{\Delta m_{\text{ice}} = 50.0 \text{ g}}$, but the final temperature of the mixture will be $T_f < 100°\text{C}$. To find the final temperature T_f, we use $Q_{\text{cold}} = -Q_{\text{hot}}$, or

$$m_{\text{ice}}L_f + m_{\text{ice}}c_w\Delta T_{\text{ice}} = -m_{\text{steam}}L_v - m_{\text{steam}}c_w\Delta T_{\text{steam}}$$

Substituting numerical values,

$$\begin{aligned}&\left(50.0\times10^{-3}\text{ kg}\right)\left(3.33\times10^{5}\text{ J/kg}\right)\\&\quad+\left(50.0\times10^{-3}\text{ kg}\right)\left(4\,186\text{ J/kg}\cdot°\text{C}\right)\left(T_f - 0°\text{C}\right)\\&\qquad=-\left(10.0\times10^{-3}\text{ kg}\right)\left(-2.26\times10^{6}\text{ J/kg}\right)\\&\qquad\quad-\left(10.0\times10^{-3}\text{ kg}\right)\left(4\,186\text{ J/kg}\cdot°\text{C}\right)\left(T_f - 100°\text{C}\right)\end{aligned}$$

From which we obtain

$$\boxed{T_f = 40.4°\text{C}}$$

(b) Since the mass of steam is much smaller than in part (a), we know that the condensation of steam will not be sufficient to melt all of the ice and raise its temperature to 100°C. We do need to determine whether the condensation of steam can supply sufficient energy to melt all of the ice. Recall from part (a) that

$$Q_1 = \left(\text{energy to melt all the ice}\right) = 1.67\times10^{4}\text{ J}$$

The energy given up as the 1.00 g of steam condenses is

$$\begin{aligned}Q_2 &= \left\{\begin{matrix}\text{energy given up}\\\text{as steam condenses}\end{matrix}\right\} = \left(10^{-3}\text{ kg}\right)\left(2.26\times10^{6}\text{ J/kg}\right)\\&= 2.26\times10^{3}\text{ J}\end{aligned}$$

Also,

$$\begin{aligned}Q_3 &= \left\{\begin{matrix}\text{energy given up as condensed}\\\text{steam cools to 0°C}\end{matrix}\right\}\\&= \left(10^{-3}\text{ kg}\right)\left(4\,186\text{ J/kg}\cdot°\text{C}\right)\left(100°\text{C}\right) = 419\text{ J}\end{aligned}$$

Since $\boxed{Q_2 + Q_3 < Q_1}$, therefore all of the steam will cool to 0°C, and $\boxed{T_f = 0°\text{C}}$ with some ice remaining. Let us now find the mass of ice which must melt to condense the steam and cool the condensate to 0°C. Again from $Q_{\text{cold}} = -Q_{\text{hot}}$,

$$m_{\text{ice}}L_f = Q_2 + Q_3 = 2.68\times10^{3}\text{ J}$$

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Thus,

$$m_{\text{ice}} = \frac{2.68\times10^{3}\ \text{J}}{3.33\times10^{5}\ \text{J/kg}} = 8.04\times10^{-3}\ \text{kg} = \boxed{8.04\ \text{g of ice melts}}$$

Therefore, there is 42.0 g of ice left over, also at 0°C.

P20.22 The boiling point of nitrogen is 77.3 K. Using units of joules, we have

$$Q = m_{\text{Cu}}c_{\text{Cu}}\Delta T = m_{\text{N}_2}\left(L_{\text{vap}}\right)_{\text{N}_2}$$

Substituting numerical values,

$$(1.00\ \text{kg})(387\ \text{J/kg}\cdot\text{°C})(293-77.3)\text{°C} = m(2.01\times10^{5}\ \text{J/kg})$$

$$m = \boxed{0.415\ \text{kg}}$$

P20.23 (a) Since the heat required to melt 250 g of ice at 0°C *exceeds* the heat required to cool 600 g of water from 18°C to 0°C, the final temperature of the system (water + ice) must be $\boxed{0\text{°C}}$.

(b) Let m represent the mass of ice that melts before the system reaches equilibrium at 0°C.

$$Q_{\text{cold}} = -Q_{\text{hot}}$$

$$mL_f = -m_w c_w\left(0\text{°C} - T_i\right)$$

$$m(3.33\times10^{5}\ \text{J/kg}) = -(0.600\ \text{kg})(4\,186\ \text{J/kg}\cdot\text{°C})(0\text{°C} - 18.0\text{°C})$$

$m = 136$ g, so the ice remaining = 250 g − 136 g = $\boxed{114\ \text{g}}$

P20.24 (a) Let n represent the number of stops. Follow the energy:

$$nK = mc\Delta T$$

$$n\left[\frac{1}{2}(1\ 500\ \text{kg})(25.0\ \text{m/s})^2\right] = (6.00\ \text{kg})(900\ \text{J/kg}\cdot\text{°C})(660\text{°C} - 20.0\text{°C})$$

$$n = \frac{3.46\times10^{6}\ \text{J}}{4.69\times10^{5}\ \text{J}} = 7.37$$

Thus $\boxed{7}$ stops can happen before melting begins.

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(b) As the car stops it transforms part of its kinetic energy into internal energy due to air resistance. As soon as the brakes rise above the air temperature they transfer energy by heat into the air, and transfer it very fast if they attain a high temperature.

Section 20.4 Work and Heat in Thermodynamic Processes

P20.25 For constant pressure, $W = -\int_{V_i}^{V_f} P dV = -P\Delta V = -P(V_f - V_i)$. Rather than evaluating the pressure numerically from atmospheric pressure plus the pressure due to the weight of the piston, we can just use the ideal gas law to write in the volumes, obtaining

$$W = -P\left(\frac{nRT_h}{P} - \frac{nRT_c}{P}\right) = -nR\left(T_h - T_c\right)$$

Therefore,

$$W = -nR\Delta T = -(0.200\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(280\ \text{K}) = \boxed{-466\ \text{J}}$$

P20.26 $$W = -\int_i^f P dV = -P\int_i^f dV = -P\Delta V = -nR\Delta T = \boxed{-nR\left(T_2 - T_1\right)}$$

The negative sign for work *on* the sample indicates that the expanding gas *does* positive work. The quantity of work is directly proportional to the quantity of gas and to the temperature change.

P20.27 During the warming process $P = \left(\frac{P_i}{V_i}\right)V$.

(a) $$W = -\int_i^f P dV = -\int_{V_i}^{3V_i} \left(\frac{P_i}{V_i}\right) V dV$$

$$W = -\left(\frac{P_i}{V_i}\right)\frac{V^2}{2}\bigg|_{V_i}^{3V_i} = -\frac{P_i}{2V_i}\left(9V_i^2 - V_i^2\right) = \boxed{-4P_iV_i}$$

(b) $PV = nRT$ gives

$$\left[\left(\frac{P_i}{V_i}\right)V\right]V = nRT \rightarrow T = \left(\frac{P_i}{nRV_i}\right)V^2$$

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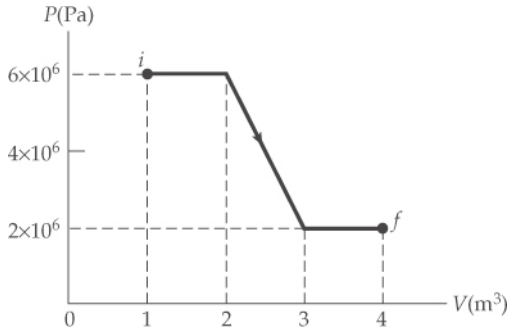
It is proportional to the square of the volume, according to $T = (P_i / nRV_i)V^2$.

P20.28 (a) $W = -\int PdV$

$$W = -\left(6.00\times10^6\ \text{Pa}\right)\left(2.00\ \text{m}^3 - 1.00\ \text{m}^3\right) +$$
$$-\left(4.00\times10^6\ \text{Pa}\right)\left(3.00\ \text{m}^3 - 2.00\ \text{m}^3\right) +$$
$$-\left(2.00\times10^6\ \text{Pa}\right)\left(4.00\ \text{m}^3 - 3.00\ \text{m}^3\right)$$

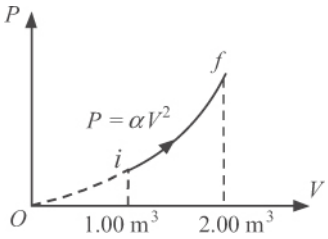
$W_{i\to f} =$ **−12.0 MJ**

(b) $W_{f\to i} =$ **+12.0 MJ**



ANS. FIG. P20.28

P20.29 The work done on the gas is the negative of the area under the curve $P = \alpha V^2$, from V_i to V_f. The work *on* the gas is negative, to mean that the expanding gas *does* positive work. We will find its amount by doing the integral



ANS. FIG. P20.29

$$W = -\int_i^f PdV$$

$$W = -\int_i^f \alpha V^2 dV = -\frac{1}{3}\alpha\left(V_f^3 - V_i^3\right)$$

$$V_f = 2V_i = 2\left(1.00\ \text{m}^3\right) = 2.00\ \text{m}^3$$

$$W = -\frac{1}{3}\left[\left(5.00\ \text{atm/m}^6\right)\left(1.013\times10^5\ \text{Pa/atm}\right)\right]$$
$$\times\left[\left(2.00\ \text{m}^3\right)^3 - \left(1.00\ \text{m}^3\right)^3\right]$$

$$= \boxed{-1.18\ \text{MJ}}$$

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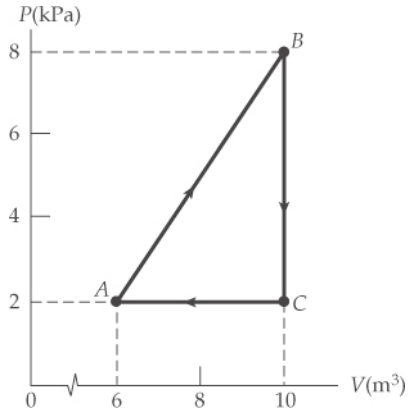
Section 20.5 The First Law of Thermodynamics

P20.30 (a) Refer to ANS. FIG. P20.30. From the first law, for a cyclic process, $Q = -W$ = Area of triangle, so

$$Q = \frac{1}{2}\left(4.00\ \text{m}^3\right)(6.00\ \text{kPa})$$

$$= \boxed{12.0\ \text{kJ}}$$

(b) $Q = -W = \boxed{-12.0\ \text{kJ}}$



ANS. FIG. P20.30

P20.31 Refer to ANS. FIG. P20.30. We tabulate the signs for Q, W, and ΔE_{int} below:

	Q	W	ΔE_{int}	
BC	–	0	–	$\left(Q = \Delta E_{\text{int}} \text{ since } W_{BC} = 0\right)$
CA	–	+	–	$\left(\Delta E_{\text{int}} < 0 \text{ and } W > 0, \text{ so } Q < 0\right)$
AB	+	–	+	$(W < 0,\ \Delta E_{\text{int}} > 0 \text{ since } \Delta E_{\text{int}} < 0$ for $B \to C \to A;$ so $Q > 0)$

P20.32 From the first law of thermodynamics,

$$\Delta E_{\text{int}} = Q + W = 10.0\ \text{J} + 12.0\ \text{J} = +22.0\ \text{J}$$

The change in internal energy is a positive number, which would be consistent with an *increase* in temperature of the gas, but the problem statement indicates a *decrease* in temperature.

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P20.33 From the first law of thermodynamics, $\Delta E_{\text{int}} = Q + W$, so

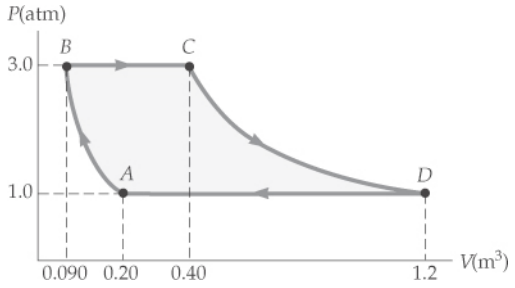
$$Q = \Delta E_{\text{int}} - W = -500\text{ J} - 220\text{ J} = \boxed{-720\text{ J}}$$

The negative sign indicates that positive energy is transferred *from* the system by heat.

P20.34 Because the gas goes through a cycle, the overall change in internal energy must be zero:

$$\Delta E_{\text{int}} = \Delta E_{\text{int},AB} + \Delta E_{\text{int},BC} + \Delta E_{\text{int},CD} + \Delta E_{\text{int},DA} = 0$$
$$\rightarrow \quad \Delta E_{\text{int},AB} = -\Delta E_{\text{int},BC} - \Delta E_{\text{int},CD} - \Delta E_{\text{int},DA}$$

Recognize that $\Delta E_{\text{int}} = 0$ for the isothermal process CD and substitute from the first law for the other internal energy changes:



ANS. FIG. P20.34

$$\begin{aligned}
\Delta E_{\text{int},AB} &= -\left(Q_{BC} + W_{BC}\right) - \left(Q_{DA} + W_{DA}\right) \\
&= -\left(Q_{BC} - P_B \Delta V_{BC}\right) - \left(Q_{DA} - P_D \Delta V_{DA}\right) \\
&= -\left(Q_{BC} + Q_{DA}\right) + \left(P_B \Delta V_{BC} + P_D \Delta V_{DA}\right) \\
&= -(345\text{ kJ} - 371\text{ kJ}) \\
&\quad + \left[(3.00\text{ atm})\left(0.310\text{ m}^3\right) + (1.00\text{ atm})\left(-1.00\text{ m}^3\right)\right] \\
&\qquad \times \left(\frac{1.013 \times 10^5\text{ Pa}}{1\text{ atm}}\right) \\
&= \boxed{4.29 \times 10^4\text{ J}}
\end{aligned}$$

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Section 20.6 Some Applications of the First Law of Thermodynamics

P20.35 (a) Rearranging $PV = nRT$ we get $V_i = \dfrac{nRT}{P_i}$

The initial volume is

$$V_i = \frac{(2.00\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(300\ \text{K})}{(0.400\ \text{atm})\left(1.013\times10^5\ \text{Pa/atm}\right)}\left(\frac{1\ \text{Pa}}{\text{N/m}^2}\right) = 0.123\ \text{m}^3$$

For isothermal compression, PV is constant, so $P_iV_i = P_fV_f$ and the final volume is

$$V_f = V_i\left(\frac{P_i}{P_f}\right) = \left(0.123\ \text{m}^3\right)\left(\frac{0.400\ \text{atm}}{1.20\ \text{atm}}\right) = \boxed{0.041\ 0\ \text{m}^3}$$

(b) $$W = -\int P\,dV = -nRT\ln\left(\frac{V_f}{V_i}\right) = -\left(4.99\times10^3\ \text{J}\right)\ln\left(\frac{1}{3}\right) = \boxed{+5.48\ \text{kJ}}$$

(c) The ideal gas keeps constant temperature so $\Delta E_{\text{int}} = 0 = Q + W$ and the heat is $Q = \boxed{-5.48\ \text{kJ}}$.

P20.36 (a) We choose as a system the H_2O molecules that all participate in the phase change. For a constant-pressure process,

$$W = -P\Delta V = -P\left(V_s - V_w\right)$$

where V_s is the volume of the steam and V_w is the volume of the liquid water. We can find them respectively from

$$PV_s = nRT \quad \text{and} \quad V_w = \text{m/p} = nM/p.$$

Calculating each work term,

$$PV_s = (1.00\ \text{mol})\left(8.314\frac{\text{J}}{\text{K}\cdot\text{mol}}\right)(373\ \text{K}) = 3\ 101\ \text{J}$$

$$PV_w = (1.00\ \text{mol})(18.0\ \text{g/mol})\left(\frac{1.013\times10^5\ \text{N/m}^2}{1.00\times10^6\ \text{g/m}^3}\right) = 1.82\ \text{J}$$

Thus the work done is

$$W = -3\ 101\ \text{J} + 1.82\ \text{J} = \boxed{-3.10\ \text{kJ}}$$

(b) The energy input by heat is

$$Q = L_v\Delta m = \left(18.0\ \text{g}\right)\left(2.26\times10^6\ \text{J/kg}\right) = 40.7\ \text{kJ}$$

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so the change in internal energy is

$$\Delta E_{\text{int}} = Q + W = 40.7 \text{ kJ} - 3.10 \text{ kJ} = \boxed{37.6 \text{ kJ}}$$

P20.37 (a) We use the energy version of the nonisolated system model.

$$\Delta E_{\text{int}} = Q + W$$

where $W = -P\Delta V$ for a constant-pressure process so that

$$\Delta E_{\text{int}} = Q - P\Delta V$$

$$= 12.5 \text{ kJ} - 2.50 \text{ kPa}\left(3.00 \text{ m}^3 - 1.00 \text{ m}^3\right) = \boxed{7.50 \text{ kJ}}$$

(b) Since pressure and quantity of gas are constant, we have from the equation of state

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

and

$$T_2 = \frac{V_2}{V_1} T_1 = \left(\frac{3.00 \text{ m}^3}{1.00 \text{ m}^3}\right)(300 \text{ K}) = \boxed{900 \text{ K}}$$

P20.38 (a) $$W = -nRT \ln\left(\frac{V_f}{V_i}\right) = -P_f V_f \ln\left(\frac{V_f}{V_i}\right)$$

Suppressing units,

$$V_i = V_f \exp\left(+\frac{W}{P_f V_f}\right) = (0.025\,0)\exp\left[\frac{-3\,000}{0.025\,0\left(1.013 \times 10^5\right)}\right]$$

$$= \boxed{0.007\,65 \text{ m}^3}$$

(b) $$T_f = \frac{P_f V_f}{nR} = \frac{1.013 \times 10^5 \text{ Pa}\left(0.025\,0 \text{ m}^3\right)}{1.00 \text{ mol}\left(8.314 \text{ J/K} \cdot \text{mol}\right)} = \boxed{305 \text{ K}}$$

P20.39 (a) $$W = -P\Delta V = -P\left[3\alpha V \Delta T\right]$$

$$= -\left(1.013 \times 10^5 \text{ N/m}^2\right)$$

$$\times \left[3\left(24.0 \times 10^{-6} (^\circ\text{C})^{-1}\right)\left(\frac{1.00 \text{ kg}}{2.70 \times 10^3 \text{ kg/m}^3}\right)(18.0^\circ\text{C})\right]$$

$$W = \boxed{-0.048\,6 \text{ J}}$$

(b) $$Q = cm\Delta T = \left(900 \text{ J/kg} \cdot ^\circ\text{C}\right)\left(1.00 \text{ kg}\right)(18.0^\circ\text{C}) = \boxed{16.2 \text{ kJ}}$$

(c) $$\Delta E_{\text{int}} = Q + W = 16.2 \text{ kJ} - 48.6 \text{ mJ} = \boxed{16.2 \text{ kJ}}$$

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P20.40 From conservation of energy, $\Delta E_{\text{int, } ABC} = \Delta E_{\text{int, } AC}$.

(a) From the first law of thermodynamics, we have

$$\Delta E_{\text{int, } ABC} = Q_{ABC} + W_{ABC}$$

Then,

$$Q_{ABC} = \Delta E_{\text{int, } ABC} - W_{ABC} = 800\text{ J} + 500\text{ J} = \boxed{1\,300\text{ J}}$$

(b) $W_{CD} = -P_C \Delta V_{CD}$, $\Delta V_{AB} = -\Delta V_{CD}$, and $P_A = 5P_C$

Then, $W_{CD} = \frac{1}{5} P_A \Delta V_{AB} = -\frac{1}{5} W_{AB} = \boxed{100\text{ J}}$.

(+ means that work is done on the system)

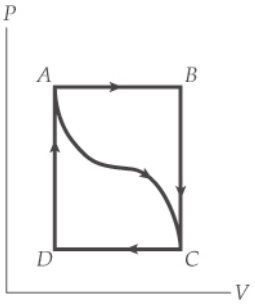
(c) $W_{CDA} = W_{CD}$ so that

$$Q_{CA} = \Delta E_{\text{int, } CA} - W_{CDA} = -800\text{ J} - 100\text{ J} = \boxed{-900\text{ J}}$$

(– means that energy must be removed from the system by heat)

(d) $\Delta E_{\text{int, } CD} = \Delta E_{\text{int, } CDA} - \Delta E_{\text{int, } DA} = -800\text{ J} - 500\text{ J} = -1\,300\text{ J}$

and $Q_{CD} = \Delta E_{\text{int, } CD} - W_{CD} = -1\,300\text{ J} - 100\text{ J} = \boxed{-1\,400\text{ J}}$.



ANS. FIG. P20.40

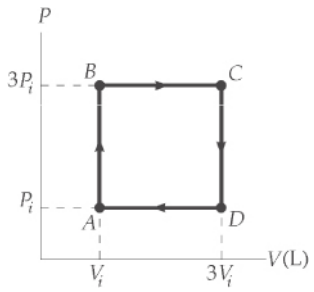
P20.41 (a) The work done during each step of the cycle equals the negative of the area under that segment of the PV curve.

$$W = W_{AB} + W_{BC} + W_{CD} + W_{DA}$$
$$W = 0 - 3P_i\left(3V_i - V_i\right) + 0 - P_i\left(V_i - 3V_i\right) + 0$$
$$W = -4P_iV_i = -4nRT_i$$
$$W = -4\left(1.00\text{ mol}\right)\left(8.314\text{ J/mol}\cdot\text{K}\right)\left(273\text{ K}\right) = \boxed{9.08\text{ kJ}}$$

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(b) The initial and final values of T for the system are equal.

Therefore, $\Delta E_{\text{int}} = 0$ and $Q = -W = \boxed{9.08 \text{ kJ}}$.



ANS. FIG. P20.41

P20.42 (a) The work done during each step of the cycle equals the negative of the area under that segment of the PV curve shown in ANS. FIG. P20.41.

$$W = W_{AB} + W_{BC} + W_{CD} + W_{DA}$$

$$W = 0 - 3P_i\left(3V_i - V_i\right) + 0 - P_i\left(V_i - 3V_i\right) + 0 = \boxed{-4P_iV_i}$$

(b) The initial and final values of T for the system are equal.

Therefore, $\Delta E_{\text{int}} = 0$ and $Q = -W = \boxed{4P_iV_i}$.

Section 20.7 Energy Transfer Mechanisms in Thermal Processes

P20.43 (a) The rate of energy transfer by conduction through a material of area A, thickness L, with thermal conductivity k, and temperatures $T_h > T_c$ on opposite sides is $P = kA\,(T_h - T_c)/L$. For the given windowpane, this is

$$P = \left(0.8 \text{ W/m}\cdot{}^\circ\text{C}\right)\left[(1.0 \text{ m})(2.0 \text{ m})\right]\frac{(25.0^\circ\text{C} - 0^\circ\text{C})}{0.620\times10^{-2} \text{ m}}$$

$$= \boxed{6.45\times10^3 \text{ W}}$$

(b) The total energy lost per day is

$$E = P\cdot\Delta t = \left(6.45\times10^3 \text{ J/s}\right)\left(8.64\times10^4 \text{ s}\right) = \boxed{5.57\times10^8 \text{ J}}$$

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P20.44 The thermal conductivity of concrete is $k = 1.3\ \text{J/s} \cdot \text{m} \cdot {}^\circ\text{C}$, so the energy transfer rate through the slab is

$$P = kA\frac{(T_h - T_c)}{L} = (0.8\ \text{W/m} \cdot {}^\circ\text{C})(5.00\ \text{m}^2)\frac{(20^\circ\text{C})}{12.0 \times 10^{-2}\ \text{m}}$$
$$= \boxed{667\ \text{W}}$$

P20.45 The net rate of energy transfer from his skin is

$$P_{\text{net}} = \sigma Ae(T^4 - T_0^4)$$
$$= (5.67 \times 10^{-8}\ \text{W/m}^2 \cdot \text{K}^4)(1.50\ \text{m}^2)$$
$$\times (0.900)\left[(308\ \text{K})^4 - (293\ \text{K})^4\right] = 125\ \text{W}$$

Note that the temperatures must be in kelvins. The energy loss in ten minutes is

$$T_{ER} = P_{\text{net}}\Delta t = (125\ \text{J/s})(600\ \text{s}) = \boxed{74.8\ \text{kJ}}$$

In the infrared, the person shines brighter than a hundred-watt light bulb.

P20.46 We find the power output of the Sun from Equation 20.19, Stefan's law:

$$P = \sigma AeT^4$$
$$= (5.669\,6 \times 10^{-8}\ \text{W/m}^2 \cdot \text{K}^4)\left[4\pi(6.96 \times 10^8\ \text{m})^2\right]$$
$$\times (0.986)(5\,800\ \text{K})^4$$
$$= \boxed{3.85 \times 10^{26}\ \text{W}}$$

P20.47 From Stefan's law,

$$P = \sigma AeT^4$$
$$2.00\ \text{W} = (5.67 \times 10^{-8}\ \text{W/m}^2 \cdot \text{K}^4)(0.250 \times 10^{-6}\ \text{m}^2)(0.950)T^4$$
$$T = (1.49 \times 10^{14}\ \text{K}^4)^{1/4} = \boxed{3.49 \times 10^3\ \text{K}}$$

P20.48 We suppose the Earth below is an insulator. The square meter must radiate in the infrared as much energy as it absorbs, $P = \sigma AeT^4$. Assuming that $e = 1.00$ for blackbody blacktop:

$$1\,000\ \text{W} = (5.67 \times 10^{-8}\ \text{W/m}^2 \cdot \text{K}^4)(1.00\ \text{m}^2)(1.00)T^4$$
$$T = (1.76 \times 10^{10}\ \text{K}^4)^{1/4} = \boxed{364\ \text{K}}$$ (You *can* cook an egg on it.)

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P20.49 (a) Because the bulb is evacuated, the filament loses energy by radiation but not by convection; we ignore energy loss by conduction. We convert the temperatures given in Celsius to Kelvin, with $T_h = 2\,100°\text{C} = 2\,373\text{ K}$ and $T_c = 2\,000°\text{C} = 2\,273\text{ K}$. Then, from Stefan's law, the power ratio is

$$e\sigma A T_h^4 / e\sigma A T_c^4 = (2\,373/2\,273)^4 = \boxed{1.19}$$

(b) The radiating area is the lateral surface area of the cylindrical filament, $2\pi r\ell$. Now we want

$$e\sigma 2\pi r_h \ell T_h^4 = e\sigma 2\pi r_c \ell T_c^4$$

so $r_c / r_h = \boxed{1.19}$

***P20.50** We use Equation 20.16 for the rate of energy transfer by conduction:

$$P = kA\frac{(T_h - T_c)}{L} = (0.210\text{ W/m}\cdot°\text{C})(1.40\text{ m}^2)\left(\frac{37.0°\text{C} - 34.0°\text{C}}{0.025\,0\text{ m}}\right)$$

$$= 35.3\text{ W} = (35.3\text{ J/s})\left(\frac{1\text{ kcal}}{4\,186\text{ J}}\right)\left(\frac{3\,600\text{ s}}{1\text{ h}}\right) = \boxed{30.3\text{ kcal/h}}$$

Since this is much less than 240 kcal/h, blood flow is essential to cool the body.

***P20.51** When the temperature of the junction stabilizes, the energy transfer rate must be the same for each of the rods, or $P_{\text{Cu}} = P_{\text{Al}}$. The cross-sectional areas of the rods are equal, and if the temperature of the junction is 50.0°C, the temperature difference is $\Delta T = 50.0°\text{C}$ for each rod. Thus,

$$P_{\text{Cu}} = k_{\text{Cu}} A\left(\frac{\Delta T}{L_{\text{Cu}}}\right) = k_{\text{Al}} A\left(\frac{\Delta T}{L_{\text{Al}}}\right) = P_{\text{Al}}$$

which gives

$$L_{\text{Al}} = \left(\frac{k_{\text{Al}}}{k_{\text{Cu}}}\right) L_{\text{Cu}} = \left(\frac{238\text{ W/m}\cdot°\text{C}}{397\text{ W/m}\cdot°\text{C}}\right)(15.0\text{ cm}) = \boxed{9.00\text{ cm}}$$

***P20.52** From $P = kA\dfrac{\Delta T}{L}$, we have

$$k = \frac{PL}{A\Delta T} = \frac{(10.0\text{ W})(0.040\,0\text{ m})}{(1.20\text{ m}^2)(15.0°\text{C})} = \boxed{2.22\times10^{-2}\text{ W/m}\cdot°\text{C}}$$

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P20.53 (a) The *R*-value of the window is the sum of the *R*-values for the two 0.125-in window panes, which Table 20.4 lists as 0.890, plus the layer of air in between. Since the Table 20.4 lists the *R*-value for an air space of 3.50 in, the total *R*-value becomes

$$R = \left[0.890 + \left(\frac{0.250}{3.50}\right)1.01 + 0.890\right]\left(\frac{\text{ft}^2 \cdot \text{°F} \cdot \text{h}}{\text{Btu}}\right)$$

$$= \boxed{1.85\ \frac{\text{ft}^2 \cdot \text{°F} \cdot \text{h}}{\text{Btu}}}$$

(b) Since A and $(T_2 - T_1)$ are constants, heat flow is reduced by a factor of $\frac{1.85}{0.890} = \boxed{2.08}$.

P20.54 (a) Intensity is defined as power per area perpendicular to the direction of energy flow. The direction of sunlight is along the line from the Sun to the object. The perpendicular area is the projected flat circular area enclosed by the *terminator*—the line that separates day from night on the object. The object radiates infrared light outward in all directions. The area perpendicular to this energy flow is its spherical surface area.

(b) The sphere of radius R absorbs sunlight over area πR^2. It radiates over area $4\pi R^2$. Then, in steady state,

$$P_{\text{in}} = P_{\text{out}}$$

$$e\left(1\,370\ \text{W/m}^2\right)\pi R^2 = e\sigma\left(4\pi R^2\right)T^4$$

The emissivity e, the radius R, and π all cancel. Therefore,

$$T = \left[\frac{1\,370\ \text{W/m}^2}{4\left(5.67 \times 10^{-8}\ \text{W/m}^2 \cdot \text{K}^4\right)}\right]^{1/4} = \boxed{279\ \text{K}} = 6\text{°C}$$

It is chilly, well below temperatures we find comfortable.

P20.55 Call the gold bar Object 1 and the silver bar Object 2. Each is a nonisolated system in steady state. When energy transfer by heat reaches a steady state, the flow rate through each will be the same, so that the junction can stay at constant temperature thereafter, with as much heat coming in through the gold as goes out through the silver.

$$P_1 = P_2 \qquad \text{or} \qquad \frac{k_1 A_1 \Delta T_1}{L_1} = \frac{k_2 A_2 \Delta T_2}{L_2}$$

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In this case, $L_1 = L_2$ and $A_1 = A_2$, so $k_1\Delta T_1 = k_2\Delta T_2$.

Let T_3 be the temperature at the junction; then

$$k_1(80.0°\text{C} - T_3) = k_2(T_3 - 30.0°\text{C})$$

Rearranging, we find

$$T_3 = \frac{(80.0°\text{C})\,k_1 + (30.0°\text{C})k_2}{k_1 + k_2}$$

$$T_3 = \frac{(80.0°\text{C})(314\ \text{W/m}\cdot°\text{C}) + (30.0°\text{C})(427\ \text{W/m}\cdot°\text{C})}{314\ \text{W/m}\cdot°\text{C} + 427\ \text{W/m}\cdot°\text{C}}$$

$$= \boxed{51.2°\text{C}}$$

P20.56 (a) The heat leaving the box during the day is given by

$$P = kA\frac{(T_H - T_c)}{L} = \frac{Q}{\Delta t}$$

$$Q = \left(0.012\ 0\frac{\text{W}}{\text{m}°\text{C}}\right)\left(0.490\ \text{m}^2\right)\left(\frac{37.0°\text{C} - 23.0°\text{C}}{0.045\ 0\ \text{m}}\right) \times (12\ \text{h})\left(\frac{3\ 600\ \text{s}}{1\ \text{h}}\right)$$

$$= 7.90 \times 10^4\ \text{J}$$

The heat lost at night is

$$Q = \left(0.012\ 0\frac{\text{W}}{\text{m}°\text{C}}\right)\left(0.490\ \text{m}^2\right)\left(\frac{37.0°\text{C} - 16.0°\text{C}}{0.045\ 0\ \text{m}}\right) \times (12\ \text{h})\left(\frac{3\ 600\ \text{s}}{1\ \text{h}}\right)$$

$$= 1.19 \times 10^5\ \text{J}$$

The total heat is $1.19 \times 10^5\ \text{J} + 7.90 \times 10^4\ \text{J} = 1.98 \times 10^5\ \text{J}$. It must be supplied by the solidifying wax: $Q = mL$

$$m = \frac{Q}{L} = \frac{1.98 \times 10^5\ \text{J}}{205 \times 10^3\ \text{J/kg}} = \boxed{0.964\ \text{kg or more}}$$

(b) The test samples and the inner surface of the insulation can be prewarmed to 37.0°C as the box is assembled. Then nothing changes in temperature during the test period and the masses of the test samples and insulation make no difference.

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P20.57 (a) Suppose the pizza is 60 cm in diameter and $\ell = 2.0 \text{ cm}$ thick, sizzling at 100°C. It cannot transfer energy by conduction or convection. It radiates according to $P = \sigma A e T^4$. Here, A is its surface area, given by

$$A = 2\pi r^2 + 2\pi r\ell = 2\pi(0.30 \text{ m})^2 + 2\pi(0.30 \text{ m})(0.02 \text{ m})$$
$$= 0.60 \text{ m}^2$$

Suppose it is dark in the infrared, with emissivity about 0.8. Then

$$P = \left(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4\right)\left(0.60 \text{ m}^2\right)(0.80)(373 \text{ K})^4$$
$$= 530 \text{ W} \boxed{\sim 10^3 \text{ W}}$$

(b) If the density of the pizza is half that of water, its mass is

$$m = \rho V = \rho\pi r^2\ell = \left(500 \text{ kg/m}^3\right)\pi(0.30 \text{ m})^2(0.02 \text{ m}) = 2.8 \text{ kg}$$

There's a lot of water in the cheese, but a lot of air in the crust, so we estimate a specific heat for the pizza between that of water and that of air. Suppose its specific heat is $c = 3\,000 \text{ J/kg} \cdot °\text{C}$. The drop in temperature of the pizza is described by

$$T_{ER} = mc\left(T_f - T_i\right)$$
$$P = \frac{dT_{ER}}{dt} = mc\frac{dT_f}{dt} - 0$$
$$\frac{dT_f}{dt} = \frac{P}{mc} = \frac{530 \text{ J/s}}{(2.8 \text{ kg})(3000 \text{ J/kg} \cdot °\text{C})}$$
$$= 0.063 \text{ °C/s} \boxed{\sim 10^{-1} \text{ K/s}}$$

Additional Problems

***P20.58** (a) Along the direct path *IF* (ANS. FIG. P20.58), the work done on the gas is the negative of the area under the curve, or

$$W = -\left[(1.00 \text{ atm})(4.00 \text{ L} - 2.00 \text{ L}) + \frac{1}{2}(4.00 \text{ atm} - 1.00 \text{ atm})(4.00 \text{ L} - 2.00 \text{ L})\right]$$
$$= (-5.00 \text{ atm} \cdot \text{L})\left(\frac{1.013 \times 10^5 \text{ Pa}}{1 \text{ atm}}\right)\left(\frac{10^{-3} \text{ m}^3}{1 \text{ L}}\right) = -507 \text{ J}$$

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Thus,

$$\Delta U = Q + W = 418\text{ J} - 507\text{ J} = \boxed{-88.5\text{ J}}$$

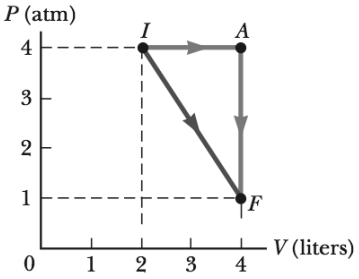
(b) Along path *IAF*, the work done on the gas is

$$W = -(4.00\text{ atm})(4.00\text{ L} - 2.00\text{ L})\left(\frac{1.013\times10^5\text{ Pa}}{1\text{ atm}}\right)\left(\frac{10^{-3}\text{ m}^3}{1\text{ L}}\right)$$

$$= -810\text{ J}$$

From the first law,

$$Q = \Delta U - W = -88.5\text{ J} - (-810\text{ J}) = \boxed{722\text{ J}}$$



ANS. FIG. P20.58

***P20.59** The constant pressure is

$$P = (1.50\text{ atm})\left(\frac{1.013\times10^5\text{ Pa}}{1\text{ atm}}\right) = 1.52\times10^5\text{ Pa}$$

and the work done on the gas is $W = -P(\Delta V)$.

(a) Here, $\Delta V = 4.00\text{ m}^3$ and

$$W = -P(\Delta V) = -\left(1.52\times10^5\text{ Pa}\right)\left(4.00\text{ m}^3\right) = \boxed{-6.08\times10^5\text{ J}}$$

(b) In this case, $\Delta V = -3.00\text{ m}^3$, so

$$W = -P(\Delta V) = -\left(1.52\times10^5\text{ Pa}\right)\left(-3.00\text{ m}^3\right) = \boxed{4.56\times10^5\text{ J}}$$

P20.60 The mass of nitrogen vaporized in a 4.00 h period is

$$m = \frac{Q}{L_f} = \frac{P\cdot(\Delta t)}{L_f} = \frac{(25.0\text{ J/s})(4.00\text{ h})(3\,600\text{ s/h})}{2.01\times10^5\text{ J/kg}} = \boxed{1.79\text{ kg}}$$

P20.61 (a) Before conduction has time to become important, the energy lost by the rod equals the energy gained by the helium. Therefore,

$$\left(mL_v\right)_{\text{He}} = \left(mc|\Delta T|\right)_{\text{Al}}$$

or $\left(\rho V L_v\right)_{\text{He}} = \left(\rho V c|\Delta T|\right)_{\text{Al}}$,

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so $$V_{\text{He}} = \frac{\left(\rho V c|\Delta T|\right)_{\text{Al}}}{\left(\rho L_v\right)_{\text{He}}}:$$

$$V_{\text{He}} = \frac{\left(2\ 700\ \text{kg/m}^3\right)\left(6.25\times10^{-5}\ \text{m}^3\right)}{\left(125\ \text{kg/m}^3\right)\left(2.09\times10^{4}\ \text{J/kg}\right)} \times\left[\left(900\ \text{J/kg}\cdot°\text{C}\right)(295.8\ \text{K})\left(\frac{1°\text{C}}{1\ \text{K}}\right)\right]$$

$$V_{\text{He}} = \left(1.72\times10^{-2}\ \text{m}^3\right)\left(\frac{1\ \text{L}}{10^{-3}\ \text{m}^3}\right) = \boxed{17.2\ \text{liters}}$$

(b) The rate at which energy is supplied to the rod in order to maintain constant temperatures is given by

$$P = kA\left(\frac{dT}{dx}\right) = \left(3\ 100\ \text{W/m}\cdot\text{K}\right)\left(2.50\times10^{-4}\ \text{m}^2\right)\left(\frac{295.8\ \text{K}}{0.250\ \text{m}}\right)$$
$$= 917\ \text{W}$$

This power supplied to the helium will produce a "boil-off" rate of

$$\frac{P}{\rho L_v} = \frac{917\ \text{W}}{\left(125\ \text{kg/m}^3\right)\left(2.09\times10^{4}\ \text{J/kg}\right)} = 3.51\ \times10^{-4}\,\text{m}^3/\text{s}$$
$$= \boxed{0.351\ \text{L/s}}$$

P20.62 (a) Isolated system (momentum). The collision is a perfectly inelastic collision, where momentum is conserved, but kinetic energy is not (it is transformed into internal energy).

(b) Momentum is conserved; thus:

$$m_1\vec{\mathbf{v}}_1 + m_2\vec{\mathbf{v}}_2 = \left(m_1 + m_2\right)\vec{\mathbf{v}} \rightarrow \vec{\mathbf{v}} = \frac{m_1\vec{\mathbf{v}}_1 + m_2\vec{\mathbf{v}}_2}{m_1 + m_2}$$

Substituting in numerically (positive to the right):

$$\vec{\mathbf{v}} = \frac{m_1\vec{\mathbf{v}}_1 + m_2\vec{\mathbf{v}}_2}{m_1 + m_2}$$
$$= \frac{(12.0\ \text{g})(+300\ \text{m/s}) + (8.00\ \text{g})(-400\ \text{m/s})}{12.0\ \text{g} + 8.00\ \text{g}}$$
$$= +20.0\ \text{m/s}$$

The final velocity is $\boxed{20.0\ \text{m/s to the right}}$.

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(c) The initial kinetic energy is

$$K_i = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_1 v_1^2$$

$$K_i = \frac{1}{2}\left(12.0\times10^{-3}\text{ kg}\right)(300.0\text{ m/s})^2 + \frac{1}{2}\left(8.00\times10^{-3}\text{ kg}\right)(-400.0\text{ m/s})^2 = 1\,180\text{ J}$$

The final kinetic energy is:

$$K_f = \frac{1}{2}\left(m_1 + m_2\right)v^2$$

$$K_f = \frac{1}{2}\left(12.0\times10^{-3}\text{ kg} + 8.00\times10^{-3}\text{ kg}\right)(20.0\text{ m/s})^2 = 4.00\text{ J}$$

The amount of kinetic energy transformed into internal energy is

$$|\Delta K| = \left|K_f - K_i\right| = |4.00\text{ J} - 1180\text{ J}| = |-1176\text{ J}| = \boxed{1.18\times10^3\text{ J}}$$

(d) $\boxed{\text{No}}$. If this amount of heat is added to the mass of the bullets, the following amount will be needed to heat the bullets to their melting temperature:

$$Q = mc\Delta T$$
$$= \left(20.0\times10^{-3}\text{ kg}\right)\left(128\text{ J/kg}\cdot\text{°C}\right)(327.3\text{°C} - 30.0\text{°C})$$
$$= 761\text{ J}$$

At the beginning of the process, 1 176 joules are generated by the collision; therefore, the bullets will be heated to the melting point, with heat still available to start the melting process:

$$1\,176\text{ J} - 761\text{ J} = 415\text{ J}$$

Therefore, 415 J are available to melt the bullets. The amount of heat needed to melt all of the combined mass of the two bullets is:

$$Q = mL = \left(20.0\times10^{-3}\text{ kg}\right)\left(2.45\times10^4\text{ J/kg}\right) = 490\text{ J}$$

There are only 415 J available; so the lead does not entirely melt due to the collision.

(e) Because there is not enough energy available to melt all the mass of the bullets, the final temperature is the melting point of lead, $\boxed{327.3\text{°C}}$.

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(f) The total mass of the melted lead is:

$$Q = mL \rightarrow m = \frac{Q}{L} = \frac{415\text{ J}}{\left(2.45\times10^{4}\text{ J/kg}\right)} = 0.016\ 94\text{ kg} = 16.9\text{ g}$$

leaving behind 20.0 g – 16.9 g = 3.10 g of unmelted solid lead:

$$\boxed{\text{3.10 g of solid lead and 16.9 g of liquid lead}}$$

P20.63 $Q = mc\Delta T = (\rho V)c\Delta T$ so that when a constant temperature difference ΔT is maintained, the rate of adding energy to the liquid is $P = \dfrac{dQ}{dt} = \rho\left(\dfrac{dV}{dt}\right)c\Delta T = \rho Rc\Delta T$ and the specific heat of the liquid is

$$c = \frac{P}{\rho R\Delta T}$$

$$= \frac{200\text{ W}}{900\text{ kg/m}^3(2.00\text{ L/min})\left(\dfrac{1\text{ min}}{60\text{ s}}\right)(3.50°\text{C})}\left(\frac{1\text{ L}}{10^{-3}\text{ m}^3}\right)$$

$$= \boxed{1.90\times10^{3}\ \frac{\text{J}}{\text{kg}\cdot°\text{C}}}$$

P20.64 $Q = mc\Delta T = (\rho V)c\Delta T$ so that when a constant temperature difference ΔT is maintained, the rate of adding energy to the liquid is $\text{P} = \dfrac{dQ}{dt} = \rho\left(\dfrac{dV}{dt}\right)c\Delta T = \rho Rc\Delta T$ and the specific heat of the liquid is

$$c = \boxed{\frac{\text{P}}{\rho R\Delta T}}.$$

P20.65 The disk is isolated, so angular momentum is conserved by the disk system. The initial moment of inertia of the disk is

$$\frac{1}{2}MR^2 = \frac{1}{2}\rho VR^2 = \frac{1}{2}\rho\left(\pi R^2 t\right)R^2$$

$$= \frac{1}{2}\left(8\,920\text{ kg/m}^3\right)\pi(28\text{ m})^4\,1.2\text{ m}$$

$$= 1.033\times10^{10}\text{ kg}\cdot\text{m}^2$$

The rotation speeds up as the disk cools off, according to

$$I_i\omega_i = I_f\omega_f$$

$$\frac{1}{2}MR_i^2\omega_i = \frac{1}{2}MR_f^2\omega_f = \frac{1}{2}MR_i^2\left(1-\alpha|\Delta T|\right)^2\omega_f$$

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$$\omega_f = \omega_i \frac{1}{(1-\alpha|\Delta T|)^2}$$

$$= (25 \text{ rad/s}) \frac{1}{\left[1-\left(17\times10^{-6} \ (°\text{C})^{-1}\right)(830°\text{C})\right]^2}$$

$$= 25.720\ 7 \text{ rad/s}$$

(a) The kinetic energy increases by

$$\frac{1}{2}I_f\omega_f^2 - \frac{1}{2}I_i\omega_i^2 = \frac{1}{2}(I_i\omega_i)\omega_f - \frac{1}{2}I_i\omega_i^2 = \frac{1}{2}I_i\omega_i(\omega_f - \omega_i)$$

$$= \frac{1}{2}\left[1.033\times10^{10} \text{ kg}\cdot\text{m}^2(25 \text{ rad/s})\right] \times(0.720\ 7 \text{ rad/s})$$

$$= \boxed{9.31\times10^{10} \text{ J}}$$

(b) $\Delta E_{\text{int}} = mc\Delta T = 2.64\times10^7 \text{ kg}(387 \text{ J/kg}\cdot°\text{C})(20°\text{C} - 850°\text{C})$

$$= \boxed{-8.47\times10^{12} \text{ J}}$$

(c) Solve the appropriate reduction of Equation 8.2 for the energy radiated by the disk:

$$\Delta K + \Delta E_{\text{int}} = T_{\text{ER}}$$

$$T_{\text{ER}} = \Delta K + \Delta E_{\text{int}} = 9.31\times10^{10}\text{J} - 8.47\times10^{12}\text{J}$$

$$= \boxed{-8.38\times10^{12}\text{J}}$$

P20.66 (a) First, energy must be removed from the liquid water to cool it to 0°C. Next, energy must be removed from the water at 0°C to freeze it, which corresponds to a liquid-to-solid phase transition. Finally, once all the water has frozen, additional energy must be removed from the ice to cool it from 0°C to $-8.00°\text{C}$.

(b) The total energy that must be removed is

$$Q = \left|Q_{\substack{\text{cool water}\\ \text{to 0°C}}}\right| + \left|Q_{\substack{\text{freeze}\\ \text{at 0°C}}}\right| + \left|Q_{\substack{\text{cool ice}\\ \text{to }-8.00°\text{C}}}\right|$$

$$= m_w c_w \left|0°\text{C} - T_i\right| + m_w L_f + m_w c_{\text{ice}} \left|T_f - 0°\text{C}\right|$$

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or

$$Q = (75.0\times10^{-3}\text{ kg})\left[(4\,186\text{ J/kg}\cdot°\text{C})|-20.0°\text{C}| + 3.33\times10^{5}\text{ J/kg} + (2\,090\text{ J/kg}\cdot°\text{C})|-8.00°\text{C}|\right]$$

$$= 3.25\times10^{4}\text{ J} = \boxed{32.5\text{ kJ}}$$

***P20.67** (a) The energy thus far gained by the copper equals the energy lost by the silver. Your down parka is an excellent insulator.

$$Q_{\text{cold}} = -Q_{\text{hot}}$$

or $m_{\text{Cu}}c_{\text{Cu}}\left(T_f - T_i\right)_{\text{Cu}} = -m_{\text{Ag}}c_{\text{Ag}}\left(T_f - T_i\right)_{\text{Ag}}$:

$$(9.00\text{ g})(387\text{ J/kg}\cdot°\text{C})(16.0°\text{C}) = -(14.0\text{ g})(234\text{ J/kg}\cdot°\text{C})\times\left(T_f - 30.0°\text{C}\right)_{\text{Ag}}$$

$$\left(T_f - 30.0°\text{C}\right)_{\text{Ag}} = -17.0°\text{C}$$

so $T_{f,\,\text{Ag}} = \boxed{13.0°\text{C}}$.

(b) Differentiating the energy gain-and-loss equation gives

$$m_{\text{Ag}}c_{\text{Ag}}\left(\frac{dT}{dt}\right)_{\text{Ag}} = -m_{\text{Cu}}c_{\text{Cu}}\left(\frac{dT}{dt}\right)_{\text{Cu}}$$

$$\left(\frac{dT}{dt}\right)_{\text{Ag}} = -\frac{m_{\text{Cu}}c_{\text{Cu}}}{m_{\text{Ag}}c_{\text{Ag}}}\left(\frac{dT}{dt}\right)_{\text{Cu}}$$

$$= -\frac{(9.00\text{ g})(387\text{ J/kg}\cdot°\text{C})}{(14.0\text{ g})(234\text{ J/kg}\cdot°\text{C})}(+0.500°\text{C/s})$$

$$\left(\frac{dT}{dt}\right)_{\text{Ag}} = \boxed{-0.532°\text{C/s}}$$

(negative sign $\Rightarrow$ decreasing temperature)

***P20.68** (a) The chemical energy input becomes partly work output and partly internal energy. The energy flow each second is described by

$$400\text{ kcal/h} = 60\text{ J/s} + \frac{mL}{\Delta t} = (400\text{ kcal/h})\left(\frac{4\,186\text{ J}}{1\text{ kcal}}\right)\left(\frac{1\text{ h}}{3\,600\text{ s}}\right)$$

$$= 465\text{ W}$$

$$\frac{m}{\Delta t}L = 465\text{ W} - 60\text{ W} = 405\text{ J/s}$$

$$\frac{m}{\Delta t} = \left(\frac{405\text{ J/s}}{2.26\times10^{6}\text{ J/kg}}\right)\left(\frac{3\,600\text{ s}}{1\text{ h}}\right) = \boxed{0.645\text{ kg/h}}$$

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(b) The rate of fat burning is ideally $\frac{400\ \text{kcal/h}}{9\ \text{kcal/g}} = 0.044\ 4\ \text{kg/h}$. The 0.044 4 kg/h of water produced by metabolism is this fraction of the water needed for cooling: $\frac{0.044\ 4\ \text{kg/h}}{0.645\ \text{kg/h}} = \boxed{0.068\ 9} = 6.89\%$.

Moral: drink plenty of water while you exercise.

***P20.69** (a) The rate of energy conversion is given by

$$Fv = (50.0\ \text{N})(40.0\ \text{m/s}) = \boxed{2\ 000\ \text{W}}$$

(b) Energy received by each object is (1 000 W)(10 s) = 10^4 J = 2 389 cal. The specific heat of iron is 0.107 cal/g · °C, so the heat capacity of each object is $5.00 \times 10^3 \times 0.107 = 535.0\ \text{cal/°C}$.

$$\Delta T = \frac{2\ 389\ \text{cal}}{535.0\ \text{cal/°C}} = \boxed{4.47°\text{C}}$$

***P20.70** We find the quantity of water vapor in one exhaled breath.

$PV = nRT$:

$$n = \frac{PV}{RT} = \frac{(3.20 \times 10^3\ \text{N/m}^2)(0.600 \times 10^{-3}\ \text{m}^3)}{(8.314\ \text{J/mol} \cdot \text{K})(273\ \text{K} + 37°\text{C})}$$

$$= 7.45 \times 10^{-4}\ \text{mol}$$

The molar mass of water (H_2O) is M = [2(1.00)+16.0] g/mol = 0.018 0 kg/mol. The mass of water vapor exhaled in one breath is

$$m_{\text{sample}} = nM = (7.45 \times 10^{-4}\ \text{mol})(0.0180\ \text{kg/mol})$$

$$= 1.34 \times 10^{-5}\ \text{kg}$$

The energy absorbed from your body as the water evaporates can be estimated as

$$Q = mL = 1.34 \times 10^{-5}\ \text{kg}(2.26 \times 10^6\ \text{J/kg}) = 30.3\ \text{J}$$

Your rate of energy loss is

$$P = \frac{Q}{\Delta t} = \left(\frac{30.3\ \text{J}}{\text{breath}}\right)\left(\frac{22.0\ \text{breath}}{\text{min}}\right)\left(\frac{1\ \text{min}}{60\ \text{s}}\right) = \boxed{11.1\ \text{W}}$$

Note that unlike a human, a dog does not perspire. Instead, the dog pants, and maximizes energy loss through the pathway considered here.

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***P20.71** The energy conservation equation is $Q_{\text{cold}} = -Q_{\text{hot}}$, or

$$m_{\text{ice}}L_f + \left[\left(m_{\text{ice}} + m_w\right)c_w + m_{\text{cup}}c_{\text{Cu}}\right](12.0°\text{C} - 0°\text{C}) = -m_{\text{Pb}}c_{\text{Pb}}(12.0°\text{C} - 98.0°\text{C})$$

This gives

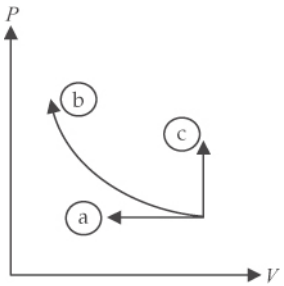
$$m_{\text{Pb}} = \left[\frac{1}{c_{\text{Pb}}(86.0°\text{C})}\right]\left\{m_{\text{ice}}L_f + \left[\left(m_{\text{ice}} + m_w\right)c_w + m_{\text{cup}}c_{\text{Cu}}\right](12.0°\text{C})\right\}$$

$$= \left[\frac{1}{(128\ \text{J/kg}\cdot°\text{C})(86.0°\text{C})}\right]\left\{(0.040\ 0\ \text{kg})\left(3.33\times10^5\ \text{J/kg}\right) + \left[(0.240\ \text{kg})(4\ 186\ \text{J/kg}\cdot°\text{C}) + (0.100\ \text{kg})(387\ \text{J/kg}\cdot°\text{C})\right](12.0°\text{C})\right\}$$

$$= \boxed{2.35\ \text{kg}}$$

P20.72 (a) Work done on the gas is the negative of the area under the PV curve:

$$W = -P_i\left(\frac{V_i}{2} - V_i\right) = \boxed{+\frac{P_iV_i}{2}}$$



ANS. FIG. P20.72

Put the cylinder into a refrigerator at absolute temperature $T_i/2$. Let the piston move freely as the gas cools.

(b) In this case the area under the curve is $W = -\int P\,dV$. Since the process is isothermal,

$$PV = P_iV_i = 4P_i\left(\frac{V_i}{4}\right) = nRT_i$$

and

$$W = -\int_{V_i}^{V_i/4}\left(\frac{dV}{V}\right)\left(P_iV_i\right) = -P_iV_i\ln\left(\frac{V_i/4}{V_i}\right) = P_iV_i\ln 4$$

$$= \boxed{+1.39P_iV_i}$$

With the gas in a constant-temperature bath at T_i, slowly push the piston in.

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(c) The area under the curve is 0 and $\boxed{W = 0}$.

Lock the piston in place and hold the cylinder over a hotplate at $3T_i$.

The student may be confused that the integral in part (c) is not explicitly covered in calculus class. Mathematicians ordinarily study integrals of functions, but the pressure is not a single-valued function of volume in a isovolumetric process. Our physics idea of an integral is more general. It still corresponds to the idea of area under the graph line.

P20.73 From Equation 8.2, for the isolated system of the meteorite and the Earth and for the time interval from when the meteorite is very far from Earth until just after it hits the Earth's surface,

$$\Delta K + \Delta U_g + \Delta E_{\text{int}} = 0 \quad \rightarrow \quad \Delta E_{\text{int}} = -\Delta K - \Delta U_g$$

The problem statement says that the internal energy increase of the system is shared equally by the meteorite and the Earth, so the change in internal energy for the meteorite *alone* is

$$\Delta E_{\text{int,meteorite}} = \frac{1}{2}\Delta E_{\text{int}} = -\frac{1}{2}\Delta K - \frac{1}{2}\Delta U_g$$

Substitute for the energies:

$$\begin{aligned}\Delta E_{\text{int,meteorite}} &= -\frac{1}{2}\left(0 - \frac{1}{2}mv_i^2\right) - \frac{1}{2}\left(-\frac{GM_E m}{R_E} - 0\right) \\ &= \frac{1}{4}mv_i^2 + \frac{GM_E m}{2R_E}\end{aligned}$$

Given the large amount of energy available for a meteorite falling to Earth, we expect the meteorite to both melt and vaporize as its temperature rises. Therefore, the internal energy change for the meteorite can be expressed as

$$\begin{aligned}\Delta E_{\text{int,meteorite}} &= mc\Delta T\big|_{\text{solid}} + L_f m + mc\Delta T\big|_{\text{liquid}} + L_v m + mc\Delta T\big|_{\text{gas}} \\ &= m\left(c\Delta T\big|_{\text{solid}} + L_f + c\Delta T\big|_{\text{liquid}} + L_v + c\Delta T\big|_{\text{gas}}\right)\end{aligned}$$

Setting the two expressions for the internal energy change of the meteorite equal gives us

$$\Delta T\big|_{\text{gas}} = \frac{\frac{1}{4}v_i^2 + \frac{GM_E}{2R_E} - c\Delta T\big|_{\text{solid}} - L_f - c\Delta T\big|_{\text{liquid}} - L_v}{c_{\text{gas}}}$$

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Substitute numerical values:

$$\Delta T\big|_{\text{gas}} = \left(\frac{1}{1170\ \text{J/kg}\cdot{}^\circ\text{C}}\right) \times\left[\frac{1}{4}\left(1.40\times10^{4}\ \text{m/s}\right)^2 + \frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)}{2\left(6.37\times10^{6}\ \text{m}\right)} - \left(900\ \text{J/kg}\cdot{}^\circ\text{C}\right)(660^\circ\text{C} + 15.0^\circ\text{C}) - 3.97\times10^{5}\ \text{J/kg} - \left(1\ 170\ \text{J/kg}\cdot{}^\circ\text{C}\right)(2\ 450^\circ\text{C} - 660^\circ\text{C}) - 1.14\times10^{7}\ \text{J/kg}\right]$$

$$= 56\ 247^\circ\text{C}$$

This is the change in temperature from the boiling point of aluminum, so to find the final temperature, add 2 450°C and express to three significant figures:

$$T_f = 56\ 247^\circ\text{C} + 2\ 450^\circ\text{C} = \boxed{5.87\times10^{4}\,^\circ\text{C}}$$

P20.74 The time interval to boil the water is related to the solar power P absorbed by the water and the energy transfer T_{ER} required:

$$\Delta t = \frac{T_{\text{ER}}}{P}$$

The energy transfer required is

$$T_{\text{ER}} = mc\Delta T$$

The solar power is

$$P = fIA = fI\left(\pi\frac{d^2}{4}\right) = \frac{1}{4}\pi d^2 fI$$

Combining all three equations,

$$\Delta t = \frac{mc\Delta T}{\left(\frac{1}{4}\pi d^2 fI\right)} = \frac{4mc\Delta T}{\pi d^2 fI}$$

$$= \frac{4\left(1.50\ \text{kg}\right)\left(4\ 186\ \text{J/kg}\cdot{}^\circ\text{C}\right)(80.0^\circ\text{C})}{\pi\left(0.600\ \text{m}\right)^2(0.400)\left(600\ \text{W/m}^2\right)}$$

$$= 7.40\times10^{3}\ \text{s} = 2.06\ \text{h}$$

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If we include setup time and coffee brewing time, this time interval approaches 2.5 hours. In the morning, the solar intensity is not the maximum amount, which occurs later in the day. Therefore, the reduced intensity in the morning will increase the time interval further. Furthermore, we have not included the energy transfer necessary to raise the temperature of the container in which the water resides. These considerations will push the required time interval even higher, so that most of the morning is used in making coffee and there is no time left for a morning hike.

P20.75 (a) The power radiated by the quiet Sun is given by Stefan's law:

$$P = \sigma AeT^4$$
$$= \left(5.67\times10^{-8}\ \text{W/m}^2\text{K}^4\right)\left[5.1\times10^{14}\ \text{m}^2\right](0.965)\left(5\,800\ \text{K}\right)^4$$
$$= \boxed{3.16\times10^{22}\ \text{W}}$$

(b) The power output of the patch of sunspot is

$$P = \left(5.67\times10^{-8}\ \text{W/m}^2\cdot\text{K}^4\right)$$
$$\times\left\{\left[0.100\left(5.10\times10^{14}\ \text{m}^2\right)\right](0.965)\left(4\,800\ \text{K}\right)^4\right.$$
$$\left.+\left[0.900\left(5.10\times10^{14}\ \text{m}^2\right)\right](0.965)\left(5\,890\ \text{K}\right)^4\right\}$$
$$= \boxed{3.17\times10^{22}\ \text{W}}$$

(c) $\boxed{\text{This is larger than } 3.158\times10^{22}\ \text{W by } \dfrac{1.29\times10^{20}\ \text{W}}{3.16\times10^{22}\ \text{W}} = 0.408\%}$

(d) $T_{\text{avg}} = 0.100\left(4\,800\ \text{K}\right) + 0.900\left(5\,890\ \text{K}\right) = \boxed{5.78\times10^3\ \text{K}}$

P20.76 (a) The block starts with $K_i = \frac{1}{2}mv_i^2 = \frac{1}{2}\left(1.60\ \text{kg}\right)\left(2.50\ \text{m/s}\right)^2 = 5.00\ \text{J}$.

Write the appropriate reduction of Equation 8.2 for the isolated copper-ice system:

$$\Delta K + \Delta E_{\text{int}} = 0 \quad \rightarrow$$
$$\left(0 - \frac{1}{2}m_{\text{Cu}}v^2\right) + L_f\Delta m = 0 \quad \rightarrow$$
$$\Delta m = \frac{m_{\text{Cu}}v^2}{2L_f}$$

Substitute numerical values:

$$\Delta m = \frac{\left(1.60\ \text{kg}\right)\left(2.50\ \text{m/s}\right)^2}{2\left(3.33\times10^5\ \text{J/kg}\right)} = 1.50\times10^{-5}\ \text{kg} = \boxed{15.0\ \text{mg}}$$

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(b) For the block as a system: $Q = 0$ (no energy transfers by heat since there is no temperature difference), $\Delta E_{\text{int}} = 0$ (no temperature or change of state).

For the block-ice system, $\Delta E_{\text{mech}} = -\Delta K = \boxed{-5.00 \text{ J}}$.

(c) For the ice as a system: $Q = 0$ (no energy transfers by heat since there is no temperature difference), $\Delta E_{\text{int}} = \Delta m L_f = \boxed{5.00 \text{ J}}$ (change of state—some ice melts).

(d) This is basically the same system as treated in part (a), treated in the same manner:

$$K_i = 5.00 \text{ J} \quad \text{and} \quad m_{\text{ice}} = \boxed{15.0 \text{ mg}}$$

(e) For the block of ice as the system: $Q = 0$ (no energy transfers by heat since there is no temperature difference),
$\Delta E_{\text{int}} = \Delta m L_f = \boxed{5.00 \text{ J}}$ (change of state—some ice melts).

For the block-ice system, $\Delta E_{\text{mech}} = -\Delta K = \boxed{-5.00 \text{ J}}$.

(f) For the metal sheet as a system, $Q = 0$ (no temperature difference), $\boxed{\Delta E_{\text{int}} = 0}$ (no change in state or temperature).

(g) Write the appropriate reduction of Equation 8.2 for the isolated copper-copper system:

$$\Delta K + \Delta E_{\text{int}} = 0 \quad \rightarrow \quad \Delta E_{\text{int}} = -\Delta K$$

Because of the symmetry of the system, each copper slab possesses half of the internal energy change of the system:

$$\Delta E_{\text{int,copper}} = \frac{1}{2}\Delta E_{\text{int}} = -\frac{1}{2}\Delta K = -\frac{1}{2}\left(0 - \frac{1}{2}mv^2\right) = \frac{1}{4}mv^2$$

The internal energy change of the copper slab is related to its temperature change:

$$\Delta E_{\text{int,copper}} = mc\Delta T = \frac{1}{4}mv^2 \quad \rightarrow \quad \Delta T = \frac{v^2}{4c}$$

Substitute numerical values:

$$\Delta T = \frac{(2.50 \text{ m/s})^2}{4(387 \text{ J/kg} \cdot {}^\circ\text{C})} = \boxed{4.04 \times 10^{-3} \ {}^\circ\text{C}}$$

(h) For the sliding slab, $\boxed{Q = 0}$ (no temperature difference), $\boxed{\Delta E_{\text{int}} = 2.50 \text{ J}}$ (friction transfers kinetic energy into internal energy).

For the two-slab system, $\boxed{\Delta E_{\text{mech}} = -5.00 \text{ J}}$ ($\Delta K = -5.00$ J).

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(i) For the stationary slab, $\boxed{Q = 0}$ (no temperature difference),

$\boxed{\Delta E_{\text{int}} = 2.50 \text{ J}}$ (friction transfers kinetic energy into internal energy).

For each object in each situation, the general continuity equation for energy, in the form $\Delta K + \Delta E_{\text{int}} = Q$, correctly describes the relationship between energy transfers and changes in the object's energy content.

P20.77 From $Q = L_v \Delta m$, the rate of boiling is described by

$$Power = \frac{Q}{\Delta t} = \frac{L_v \Delta m}{\Delta t}$$

so that the mass flow rate of steam from the kettle is

$$\frac{\Delta m}{\Delta t} = \frac{Power}{L_v}$$

The symbols Δm for mass vaporized and m for mass leaving the kettle have the same meaning, but recall that M represents the molar mass. Even though it is on the point of liquefaction, we model the water vapor as an ideal gas. The volume flow rate $V / \Delta t$ of the fluid is the cross-sectional area of the spout multiplied by the speed of flow, forming the product Av.

$$P_0 V = nRT = \left(\frac{m}{M}\right)RT$$

$$\frac{P_0 V}{\Delta t} = \frac{m}{\Delta t}\left(\frac{RT}{M}\right)$$

$$P_0 Av = \frac{Power}{L_v}\left(\frac{RT}{M}\right)$$

$$v = \frac{(Power)RT}{ML_v P_0 A}$$

Suppressing units,

$$v = \frac{(1\ 000)(8.314)(373)}{(0.018\ 0)\left(2.26\times10^{6}\right)\left(1.013\times10^{5}\right)\left(2.00\times10^{-4}\right)} = \boxed{3.76 \text{ m/s}}$$

P20.78 $A = A_{\text{end walls}} + A_{\text{ends of attic}} + A_{\text{side walls}} + A_{\text{roof}}$

$$A = 2(8.00 \text{ m} \times 5.00 \text{ m}) + 2\left[2 \times \frac{1}{2} \times 4.00 \text{ m} \times (4.00 \text{ m}) \tan 37.0^\circ\right]$$

$$+ 2(10.0 \text{ m} \times 5.00 \text{ m}) + 2(10.0 \text{ m})\left(\frac{4.00 \text{ m}}{\cos 37.0^\circ}\right)$$

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$$A = 304 \text{ m}^2$$

$$P = \frac{kA\Delta T}{L} = \frac{(4.80\times10^{-4} \text{ kW/m}\cdot\text{°C})(304 \text{ m}^2)(25.0\text{°C})}{0.210 \text{ m}} = 17.4 \text{ kW}$$
$$= 4.15 \text{ kcal/s}$$

Thus, the energy transferred through the walls per day by heat is

$$(4.15 \text{ kcal/s})(86\,400 \text{ s}) = 3.59\times10^5 \text{ kcal/day}$$

The gas needed to replace this transfer is $\dfrac{3.59\times10^5 \text{ kcal/day}}{9\,300 \text{ kcal/m}^3}$

$$= \boxed{38.6 \text{ m}^3/\text{day}}$$

P20.79 Energy goes in at a constant rate P. For the period from 50.0 min to 60.0 min, after the ice has melted,

$$P\Delta t = Q = mc\Delta T$$

$$P(10.0 \text{ min}) = (10 \text{ kg} + m_i)(4\,186 \text{ J/kg}\cdot\text{°C})(2.00\text{°C} - 0\text{°C})$$

$$P(10.0 \text{ min}) = 83.7 \text{ kJ} + (8.37 \text{ kJ/kg})m_i \qquad \textbf{[1]}$$

For the period from 0 to 50.0 min, as the ice is melting,

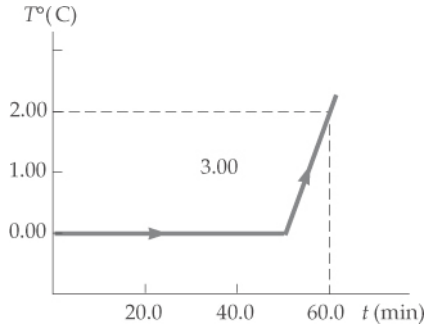
$$P\Delta t = Q = m_i L_f$$

$$P(50.0 \text{ min}) = m_i(3.33\times10^5 \text{ J/kg})$$

Substitute $P = \dfrac{m_i(3.33\times10^5 \text{ J/kg})}{50.0 \text{ min}}$ into equation [1] to find

$$\frac{m_i(3.33\times10^5 \text{ J/kg})}{5.00} = 83.7 \text{ kJ} + (8.37 \text{ kJ/kg})m_i$$

$$m_i = \frac{83.7 \text{ kJ}}{(66.6 - 8.37) \text{ kJ/kg}} = \boxed{1.44 \text{ kg}}$$



ANS. FIG. P20.79

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P20.80 (a) From conservation of energy, where the subscript w is for water and the subscript c is for the calorimeter,

$$Q_{\text{cold}} = -Q_{\text{hot}} \quad \text{or} \quad Q_{\text{Al}} = -\left(Q_w + Q_c\right)$$

$$m_{\text{Al}} c_{\text{Al}} \left(T_f - T_i\right)_{\text{Al}} = -\left(m_w c_w + m_c c_c\right)\left(T_f - T_i\right)_w$$

$$\left(0.200\ \text{kg}\right) c_{\text{Al}} \left(+39.3°\text{C}\right)$$

$$= -\left[\left(0.400\ \text{kg}\right)\left(4\,186\ \text{J/kg}\cdot°\text{C}\right) + \left(0.040\,0\ \text{kg}\right)\left(630\ \text{J/kg}\cdot°\text{C}\right)\right]\left(-3.70°\text{C}\right)$$

$$c_{\text{Al}} = \frac{6.29 \times 10^3\ \text{J}}{7.86\ \text{kg}\cdot°\text{C}} = \boxed{800\ \text{J/kg}\cdot°\text{C}}$$

(b) $$\frac{900 - 800}{900} = 11\%$$

This differs from the tabulated value by 11%, so the values agree within 15%.

Challenge Problems

P20.81 (a) The speed of rise of the piston is the same as the rate at which the height h of the steam above the water is increasing due to the boiling process. The volume of the gas is the area A of the cylinder times the height h:

$$v = \frac{dh}{dt} = \frac{d}{dt}\left(\frac{V}{A}\right)$$

The volume of steam can be replaced using the ideal gas law, in which the pressure P and temperature T are constant:

$$v = \frac{1}{A}\frac{d}{dt}\left(\frac{nRT}{P}\right) = \frac{RT}{PA}\frac{dn}{dt}$$

The combination PA is the force applied by the gas on the piston. Assuming that the speed of the piston is constant, the piston is in equilibrium so this force is equal to the product of atmospheric pressure P_0 and the area of the piston plus the weight $m_p g$ of the piston.

$$F = P_0 A + m_p g$$

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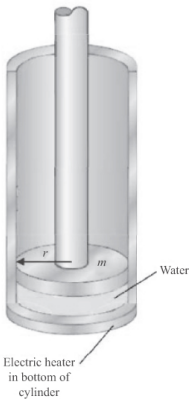
The number of moles n is the ratio of the mass m_g of the gas to the molecular mass M_w:

$$n = \left(\frac{m_g}{M_w} \right)$$

which may both be substituted into our velocity equation:

$$v = \left(\frac{RT}{m_p g + P_0 A} \right) \frac{d}{dt} \left(\frac{m_g}{M_w} \right)$$

$$= \left[\frac{RT}{\left(m_p g + P_0 A \right) M_w} \right] \frac{dm_g}{dt}$$



ANS. FIG. P20.81

The change in the mass of the steam is related to the latent heat of vaporization by Equation 20.7:

$$Q = \pm(\Delta m)L \qquad \rightarrow \frac{dm_g}{dt} = \frac{d}{dt} \left(\frac{Q}{L_v} \right)$$

$$v = \frac{RT}{\left(m_p g + P_0 A \right) M_w} \frac{d}{dt} \left(\frac{Q}{L_v} \right) = \frac{RT}{\left(m_p g + P_0 A \right) M_w L_v} \frac{dQ}{dt}$$

Finally, the rate at which energy is entering the cylinder is the power, (*Power*): (Notice that here we are careful to distinguish power from pressure P which normally would use the same symbols.)

$$v = \frac{RT(Power)}{\left(m_p g + P_0 A \right) M_w L_v}$$

Now we substitute numerical values, suppressing units:

$$v = \frac{(8.314)(373)(100)}{\left[(3.00)(9.80) + \left(1.013 \times 10^5 \right)(\pi)(0.0750)^2 \right](0.0180)\left(2.26 \times 10^6 \right)}$$

$$= 4.19 \times 10^{-3} \text{ m/s} = \boxed{4.19 \text{ mm/s}}$$

(b) Begin the same way as part (a):

$$v = \frac{dh}{dt} = \frac{d}{dt} \left(\frac{V}{A} \right) = \frac{1}{A} \frac{d}{dt} \left(\frac{nRT}{P} \right)$$

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In this situation, however, the number of moles n is fixed and the temperature T changes:

$$v = \left(\frac{nR}{PA}\right)\frac{dT}{dt} = \frac{nR}{\left(m_p g + P_0 A\right)}$$

The temperature change is related to the energy input by means of Equation 20.4:

$$Q = mc\Delta T \rightarrow \frac{dT}{dt} = \frac{d}{dt}\left(\frac{Q}{m_g c}\right)$$

which may be substituted into our velocity equation:

$$v = \frac{nR}{\left(m_p g + P_0 A\right)}\frac{d}{dt}\left(\frac{Q}{m_g c}\right) = \frac{m_g R}{\left(m_p g + P_0 A\right)m_g c M_w}\frac{dQ}{dt}$$

$$v = \frac{R(Power)}{\left(m_p g + P_0 A\right)cM_w}$$

Substitute numerical values, suppressing units,

$$v = \frac{(8.314)(100)}{\left[(3.00)(9.80)+\left(1.013\times 10^5\right)(\pi)(0.0750)^2\right](2\ 010)(0.018\ 0)}$$

$$= 0.012\ 6\ \text{m/s} = \boxed{12.6\ \text{mm/s}}$$

P20.82 (a) If the energy transfer P through one spherical surface within the shell were different from the energy transfer through another sphere, the temperature would be changing at a radius between the layers, so the steady state would not yet be established.

The equation $dT/dr = P/4\pi kr^2$ represents the law of thermal conduction, incorporating the definition of thermal conductivity, applied to a spherical surface within the shell. The rate of energy transfer P must be the same for all radii so that each bit of material stays at a temperature that is constant in time.

(b) we separate the variables T and r in the thermal conduction equation and integrate the equation between points on the interior and exterior surfaces.

$$\int_5^{40} dT = \frac{P}{4\pi k}\int_{0.03}^{0.07}\frac{dr}{r^2}$$

where T is in degrees Celsius, P is in watts, and r is in meters.

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(c) The integral yields

$$T\Big|_5^{40} = \frac{P}{4\pi k}\left(\frac{r^{-1}}{-1}\right)\Bigg|_{0.03}^{0.07}$$

$$40 - 5 = \frac{P}{4\pi(0.8)}\left(-\frac{1}{0.07} + \frac{1}{0.03}\right)$$

$$P = \boxed{18.5\ \text{W}}$$

(d) With P now known, we separate the variables again and integrate between a point on the interior surface and any point within the shell.

$$\int_5^T dT = \frac{P}{4\pi k}\int_{0.03}^{r}\frac{dr}{r^2}$$

(e) Integrating, we find

$$T\Big|_5^T = \frac{P}{4\pi k}\left(\frac{r^{-1}}{-1}\right)\Bigg|_{0.03}^{r} \quad \rightarrow \quad T - 5 = \frac{18.5}{4\pi(0.8)}\left(-\frac{1}{r} + \frac{1}{0.0300}\right)$$

$$\boxed{\begin{array}{l} T = 5 + 1.84\left(\dfrac{1}{0.030\ 0} - \dfrac{1}{r}\right) \\ \text{Where } T \text{ is in degrees Celsius and } r \text{ is in meters} \end{array}}$$

(f) $$T = 5 + 1.84\left(\frac{1}{0.0300} - \frac{1}{r}\right) = 5 + 1.84\left(\frac{1}{0.0300} - \frac{1}{0.0500}\right) = \boxed{29.5^\circ\text{C}}$$

P20.83 $$\frac{L\rho A dx}{dt} = kA\left(\frac{\Delta T}{x}\right)$$

$$L\rho\int_{4.00}^{8.00} x\,dx = k\Delta T\int_0^{\Delta t} dt$$

$$L\rho\frac{x^2}{2}\Bigg|_{4.00}^{8.00} = k\Delta T\Delta t$$

$$\left(3.33\times10^5\ \text{J/kg}\right)\left(917\ \text{kg/m}^3\right)\left(\frac{(0.080\ 0\ \text{m})^2 - (0.040\ 0\ \text{m})^2}{2}\right) = \left(2.00\ \text{W/m}\cdot^\circ\text{C}\right)(10.0^\circ\text{C})\Delta t$$

$$\Delta t = 3.66\times10^4\ \text{s} = \boxed{10.2\ \text{h}}$$

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P20.84 (a) See ANS. FIG. P20.84. For a cylindrical shell of radius r, height L, and thickness dr, the equation for thermal conduction,

$$\frac{dQ}{dt} = -kA\frac{dT}{dx} \quad \text{becomes} \quad \frac{dQ}{dt} = -k(2\pi rL)\frac{dT}{dr}$$

Under equilibrium conditions, $\frac{dQ}{dt}$ is constant; therefore,

$$dT = -\frac{dQ}{dt}\left(\frac{1}{2\pi kL}\right)\left(\frac{dr}{r}\right) \quad \text{and} \quad \int_{T_a}^{T_b} dT = -\frac{dQ}{dt}\left(\frac{1}{2\pi kL}\right)\int_a^b \frac{dr}{r}$$

$$T_b - T_a = -\frac{dQ}{dt}\left(\frac{1}{2\pi kL}\right)\ln\left(\frac{b}{a}\right)$$

But $T_a > T_b$, so $\boxed{\frac{dQ}{dt} = 2\pi kL\left[\frac{(T_a - T_b)}{\ln(b/a)}\right]}$

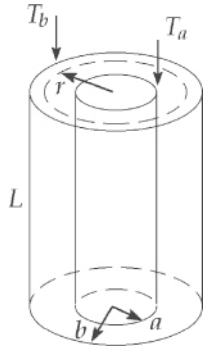
(b) From part (a), the rate of energy flow through the wall is

$$\frac{dQ}{dt} = \frac{2\pi kL(T_a - T_b)}{\ln(b/a)}$$

$$\frac{dQ}{dt} = \frac{2\pi\left(4.00\times10^{-5}\ \text{cal/s}\cdot\text{cm}\cdot{}^\circ\text{C}\right)(3\,500\ \text{cm})(60.0^\circ\text{C})}{\ln(256\ \text{cm}/250\ \text{cm})}$$

$$\frac{dQ}{dt} = 2.23\times10^3\ \text{cal/s} = \boxed{9.32\ \text{kW}}$$

This is the rate of energy loss from the plane by heat, and consequently is the rate at which energy must be supplied in order to maintain a constant temperature.



ANS. FIG. P20.84(b)

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P20.2 0.105° C

P20.4 16.9° C

P20.6 0.234 kJ/kg · °C

P20.8 87.0°C

P20.10 $\dfrac{(m_{\mathrm{Al}}c_{\mathrm{Al}} + m_c c_w)T_c + m_h c_w T_h}{m_{\mathrm{Al}}c_{\mathrm{Al}} + m_c c_w + m_h c_w}$

P20.12 (a) 16.1°C; (b) 16.1°C; (c) It makes no difference whether the drill bit is dull or sharp, or how far into the block it cuts. The answers to (a) and (b) are the same because all of the work done by the bit on the block constitutes energy being transferred into the internal energy of the steel.

P20.14 (a) T = 25.8°C; (b) The symbolic result from part (a) shows no dependence on mass. Both the change in gravitational potential energy and the change in internal energy of the system depend on the mass, so the mass cancels.

P20.16 12.9 g steam

P20.18 1.22×10^5 J

P20.20 0.294 g

P20.22 0.415 kg

P20.24 (a) 7; (b) As the car stops, it transforms part of its kinetic energy into internal energy due to air resistance. As soon as the brakes rise above the air temperature, they transfer energy by heat into the air and transfer it very fast if they attain a high temperature.

P20.26 $-nR(T_2 - T_1)$

P20.28 (a) −12.0 MJ; (b) + 12.0 MJ

P20.30 (a) 12.0 kJ; (b) −12.0 kJ

P20.32 From the first law of thermodynamics, $\Delta E_{\mathrm{int}} = Q + W$ = 10.0 J + 12.0 J = +22.0 J. The change in internal energy is a positive number, which would be consistent with an *increase* in temperature of the gas, but the problem statement indicates a *decrease* in temperature.

P20.34 4.29×10^4 J

P20.36 (a) −3.10 kJ; (b) 37.6 kJ

P20.38 (a) 0.007 65 m^3; (b) 305 K

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P20.40 (a) 1 300 J; (b) 100 J; (c) −900 J; (d) −1 400 J

P20.42 (a) $-4\,P_iV_i$; (b) $4\,P_iV_i$

P20.44 667 W

P20.46 3.85×10^{26} W

P20.48 364 K

P20.50 30.3 kcal/h

P20.52 2.22×10^{-2} W/m · °C

P20.54 (a) Intensity is defined as power per area perpendicular to the direction of energy flow. The direction of sunlight is along the line from the Sun to an object. The perpendicular area is the projected flat circular area enclosed by the *terminator*. The object radiates infrared light outward in all directions. The area perpendicular to this energy flow is its spherical surface area; (b) 279 K, it is chilly, well below room temperatures we find comfortable.

P20.56 (a) 0.964 kg or more; (b) The test samples and the inner surface of the insulation can be pre-warmed to 37.0° as the box is assembled. Then, nothing changes in temperature during the test period and the masses of the test samples and insulation make no difference.

P20.58 (a) –88.5 J; (b) 722 J

P20.60 1.79 kg

P20.62 (a) Isolated system (momentum). The collision is a perfectly inelastic collision, where momentum is conserved, but kinetic energy is not. (It is transformed to internal energy); (b) 20.0 m/s to the right; (c) 1.18×10^3 J; (d) No; (e) 327.3°C; (f) 3.1 g of solid lead and 16.9 g of liquid lead

P20.64 $\dfrac{P}{\rho R \Delta T}$

P20.66 (a) First, energy must be removed from the liquid water to cool it to 0° C. Next, energy must be removed from the water at 0° C to freeze it, which corresponds to a liquid-to-solid phase transition. Finally, once all the water has frozen, additional energy must be removed from the ice to cool it from 0° to −8.00°C; (b) 32.5 kJ

P20.68 (a) 0.645 kg/h; (b) 0.068 9

P20.70 11.1 W

P20.72 (a) $+\dfrac{P_iV_i}{2}$; Put the cylinder into a refrigerator at absolute temperature $T/2$. Let the piston move freely as the gas cools; (b) $+1.39P_iV_i$; With the gas in a constant-temperature bath at T_f, slowly push the piston in; (c)

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$W = 0$; Lock the piston in place and hold the cylinder over at hotplate at $3T_f$.

P20.74 Most of the morning is used in making coffee, and there is no time left for a morning hike.

P20.76 (a) 15.0 mg; (b) –5.00 J; (c) 5.00 J; (d) 15.0 mg; (e) 5.00 J; (f) $\Delta E_{\text{int}} = 0$; (g) 4.04×10^{-3} °C; (h) $Q = 0$; $\Delta E_{\text{int}} = 250$ J; $\Delta E_{\text{mech}} = -5.00$ J; (i) $Q = 0$; $\Delta E_{\text{int}} = 2.50$ J

P20.78 38.6 m^3/day

P20.80 (a) 800 J/kg · °C; (b) This differs from the tabulated value by 11%, so the values agree within 15%.

P20.82 (a) The equation $dT/dr = P/4\pi kr^2$ represents the law of thermal conduction, incorporating the definition of thermal conductivity, applied to a spherical surface within the shell. The rate of energy transfer P must be the same for all radii so that each bit of material stays at a temperature that is constant in time; (b) See P20.70(b) for full proof; (c) 18.5 W; (d) See P20.70(d) for full proof;

(e) $T = 5 + 1.84\left(\dfrac{1}{0.030\ 0} - \dfrac{1}{r}\right)$, where T is in degrees Celsius and r is in meters; (f) 29.5° C

P20.84 (a) $\dfrac{dQ}{dt} = 2\pi kL\left[\dfrac{(T_a - T_b)}{\ln(b/a)}\right]$; (b) 9.32 kW

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